
بسم الله الرحمن الرحيم

DISASTER

- **Widespread, varies**
  - **Spatially-Location and extent**
  - **Temporally- Repeat**
- **Resources (Rescue, Relief, Rehabilitation)**
  - **Data (constraints)**
    - **Pre and Post Events**
    - **Scale / Resolution**
    - **Availability and cost**
    - **Expertise**

Floods
Risk is a spatial problem

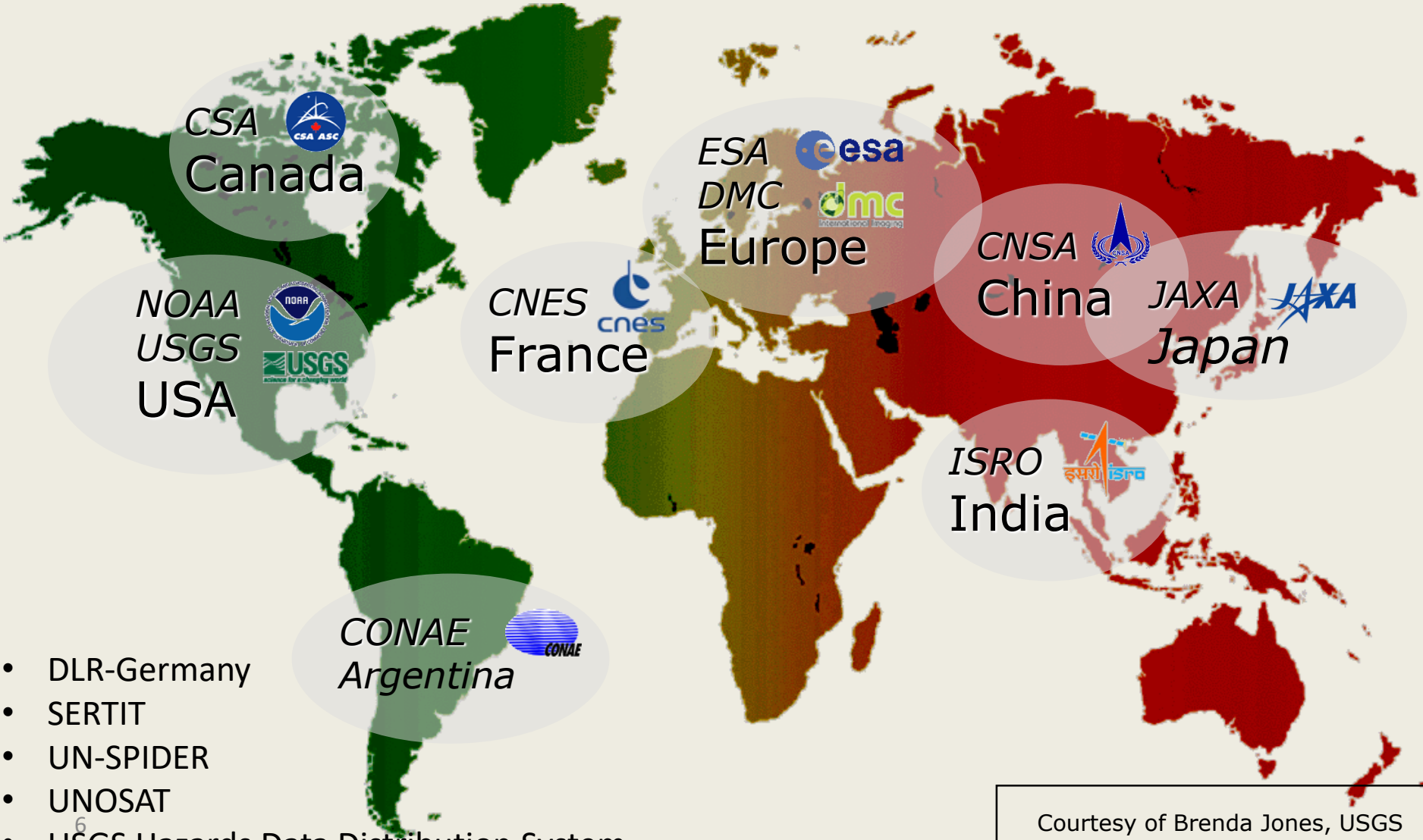
- **Hazard:**
  - How much water when and where?
- **Elements at risk:**
  - Which elements where, and how many/much ?
- **Vulnerability:**
  - How much water where which elements at risk are?

How to answer these questions?

- Where did hazards occur before?
- What type did occur?
- How often did they occur?
- Why did they happen there?
- What are the main causes?
- How can we map these?
- Where will they happen in the coming period?
- How frequent will they happen in future?
- How big will they be?
- What would be the expected damage, and where?
- What can we do to decrease this expected damage?

Charter Member Agencies

The International Charter Space and Major Disasters

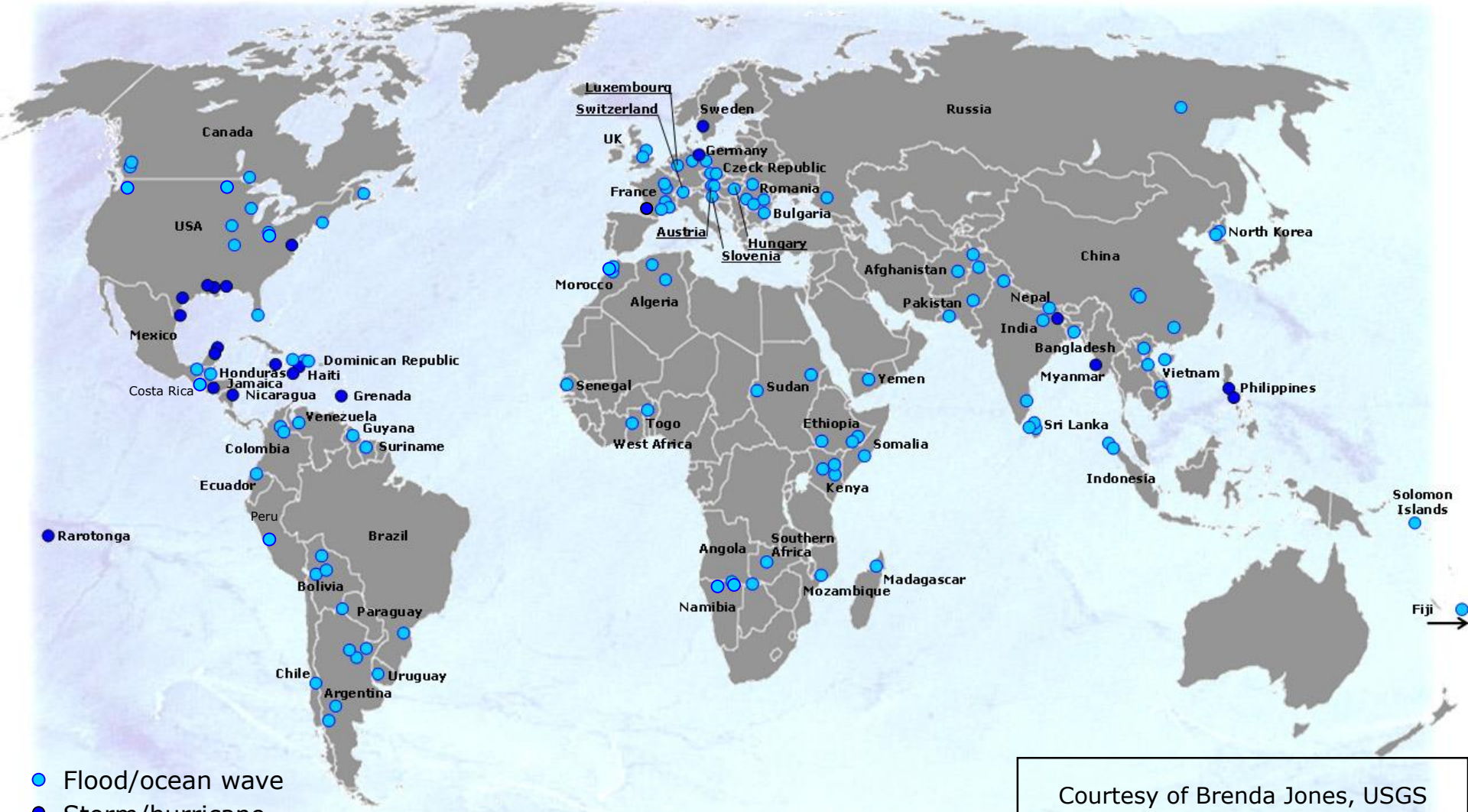


- DLR-Germany
- SERTIT
- UN-SPIDER
- UNOSAT
- USGS Hazards Data Distribution System

Courtesy of Brenda Jones, USGS

Activation Distribution

(Floods and storms)



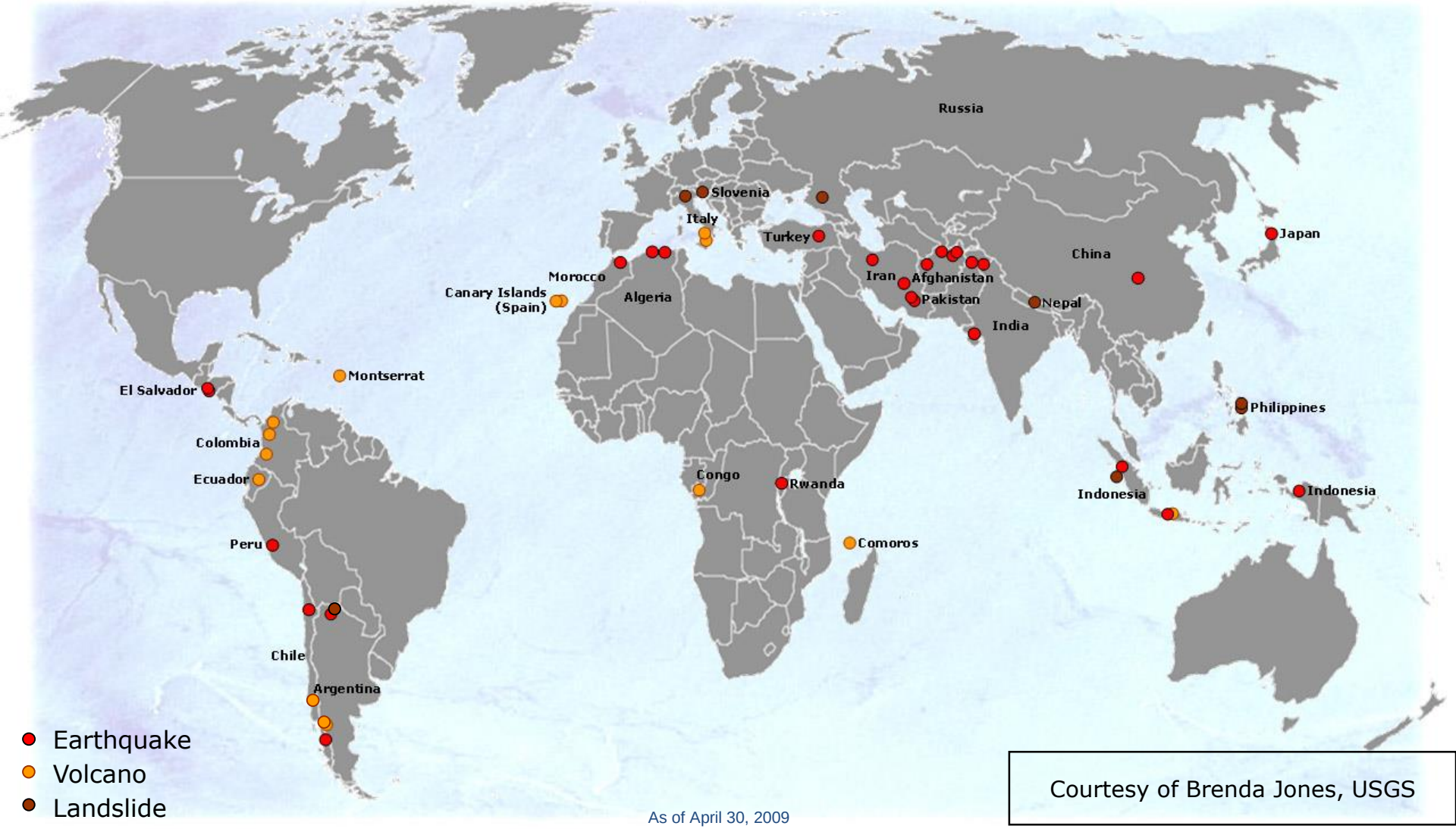
- Flood/ocean wave
- Storm/hurricane

As of April 30, 2009

Courtesy of Brenda Jones, USGS

Activation distribution

Earthquakes, volcanic eruptions and landslides

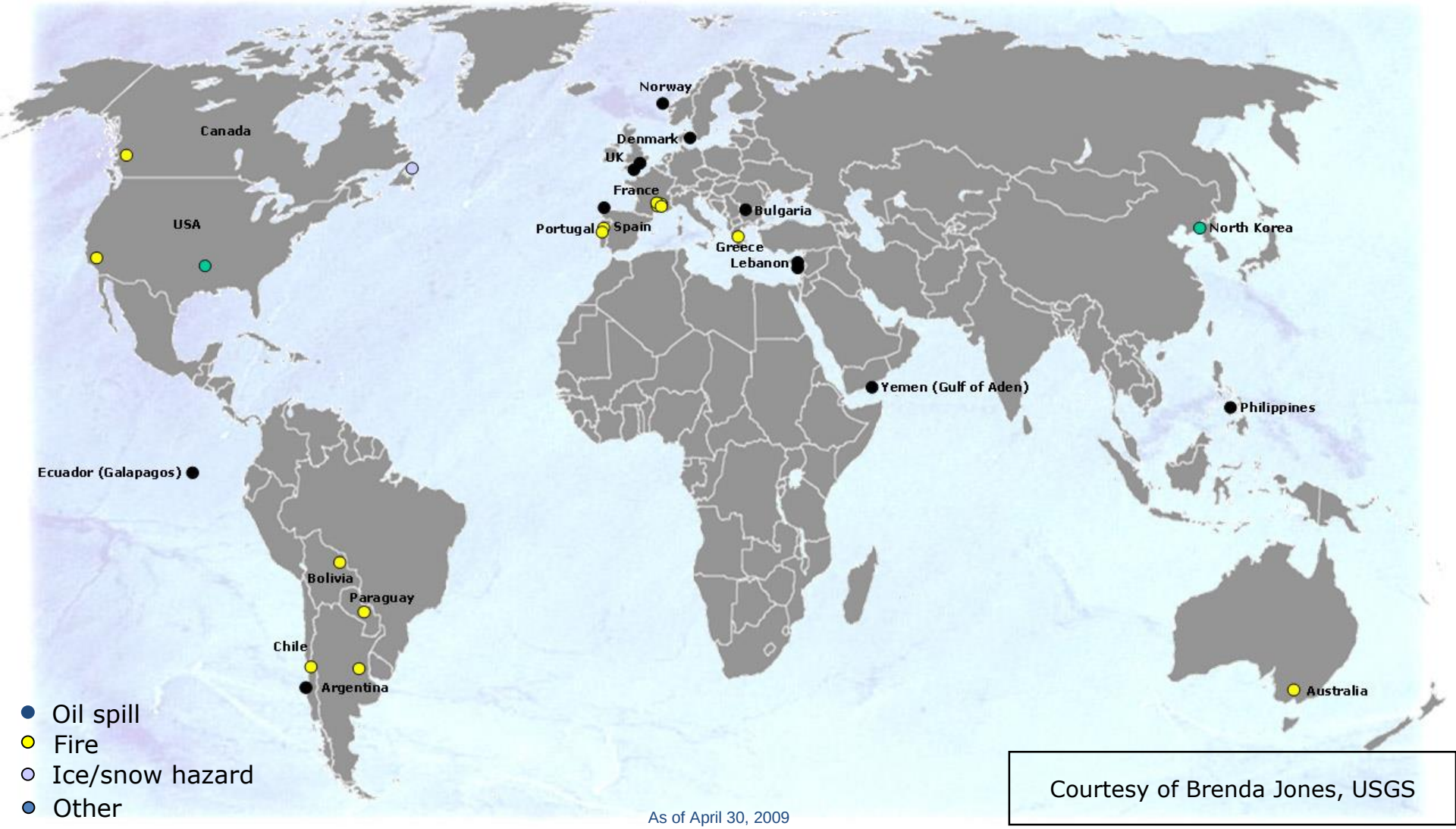


Courtesy of Brenda Jones, USGS

As of April 30, 2009

Activation distribution

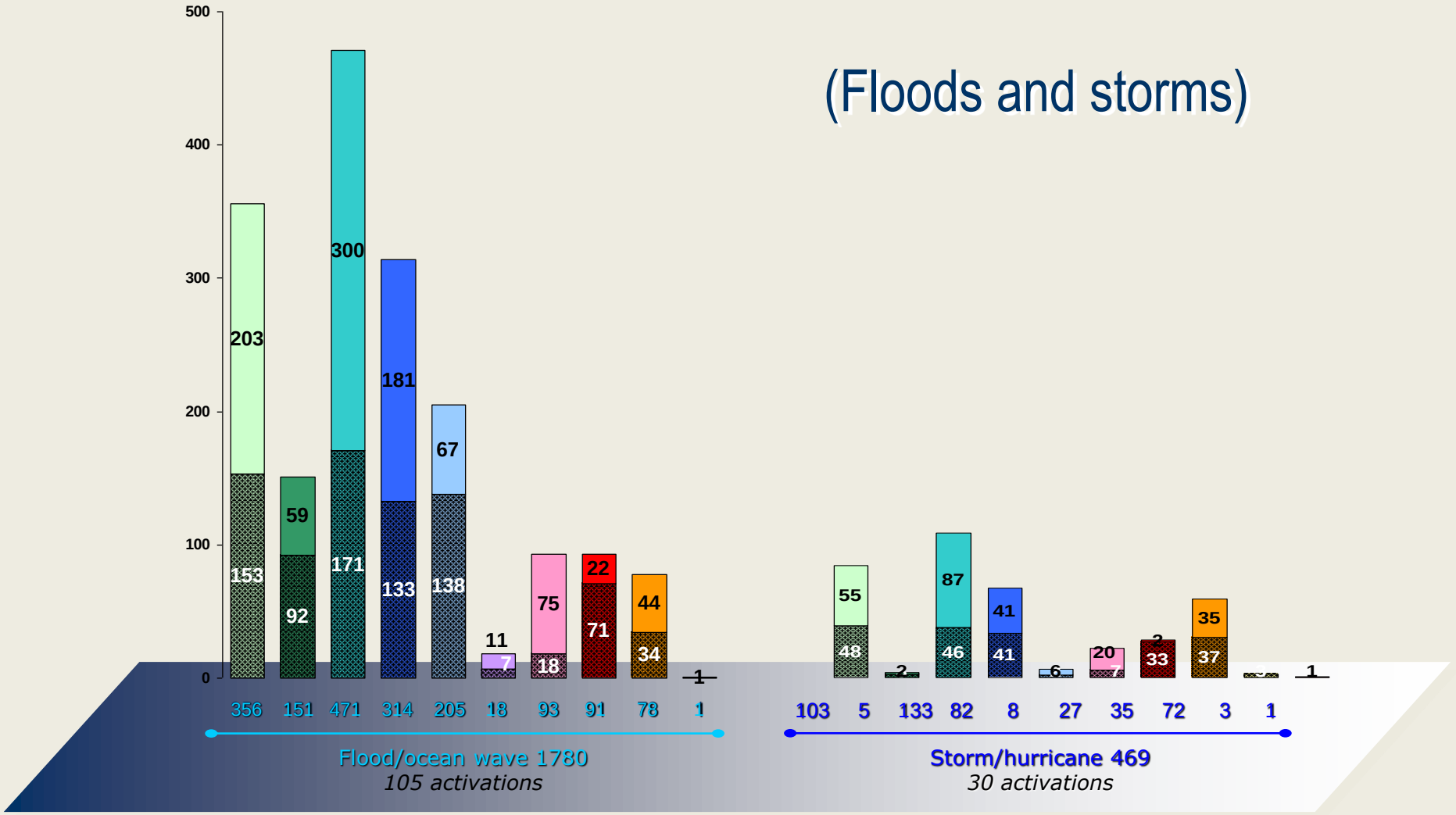
Oil spills, forest fires and other hazards

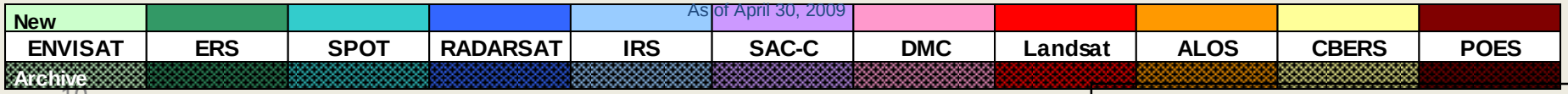


As of April 30, 2009

Courtesy of Brenda Jones, USGS

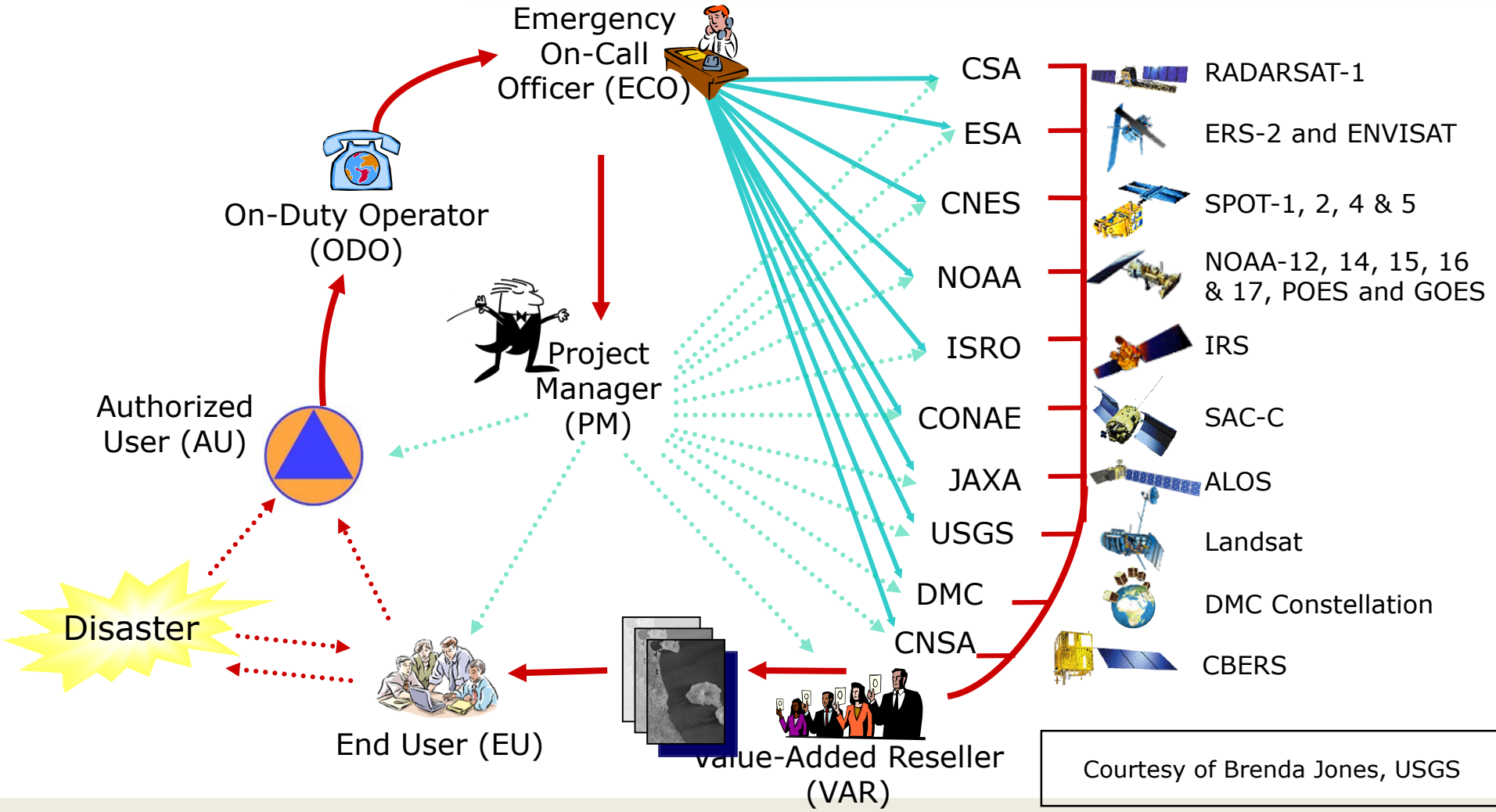
Data Units Used for Various Disasters





Courtesy of Brenda Jones, USGS

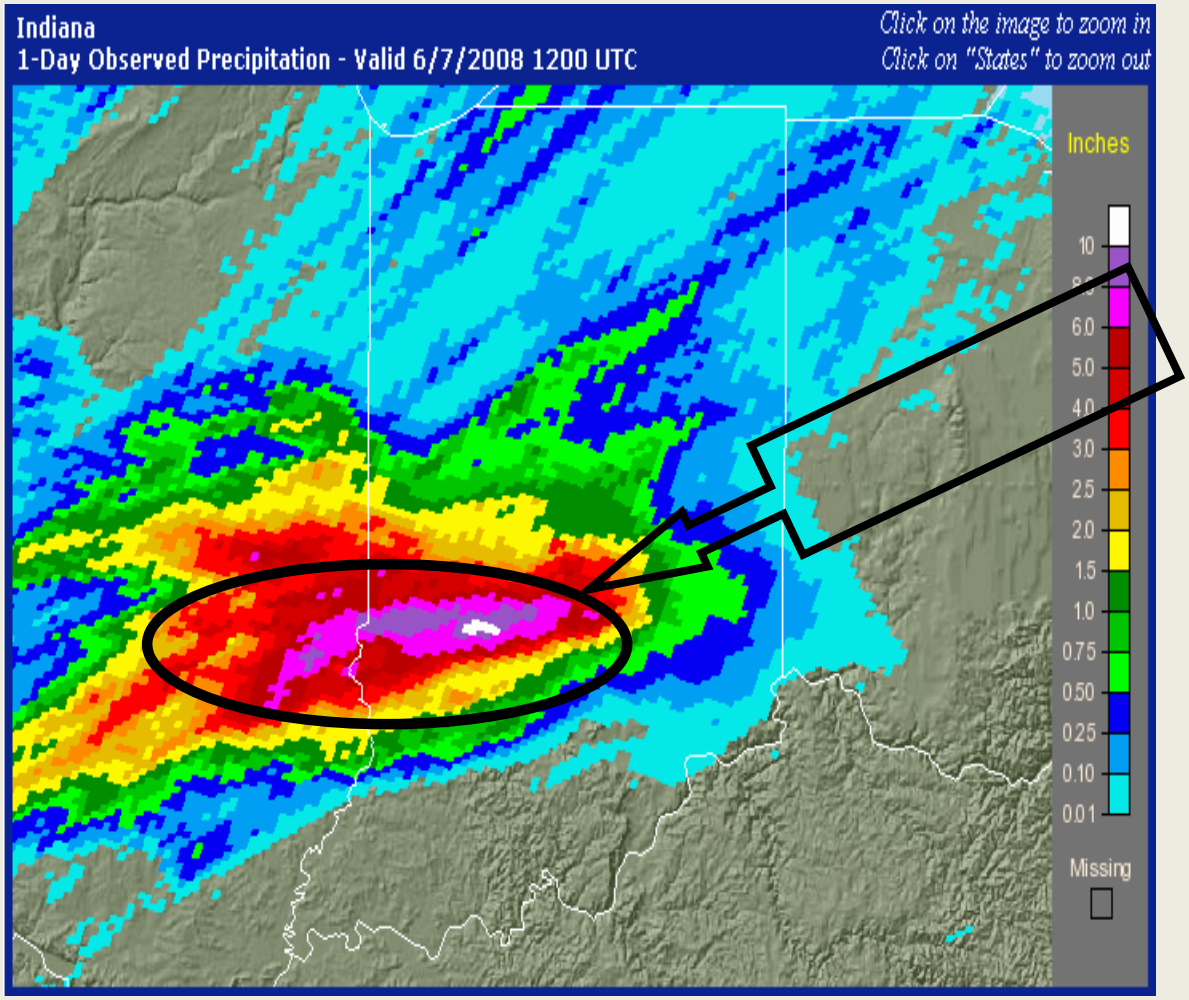
Charter Operational Loop



Courtesy of Brenda Jones, USGS

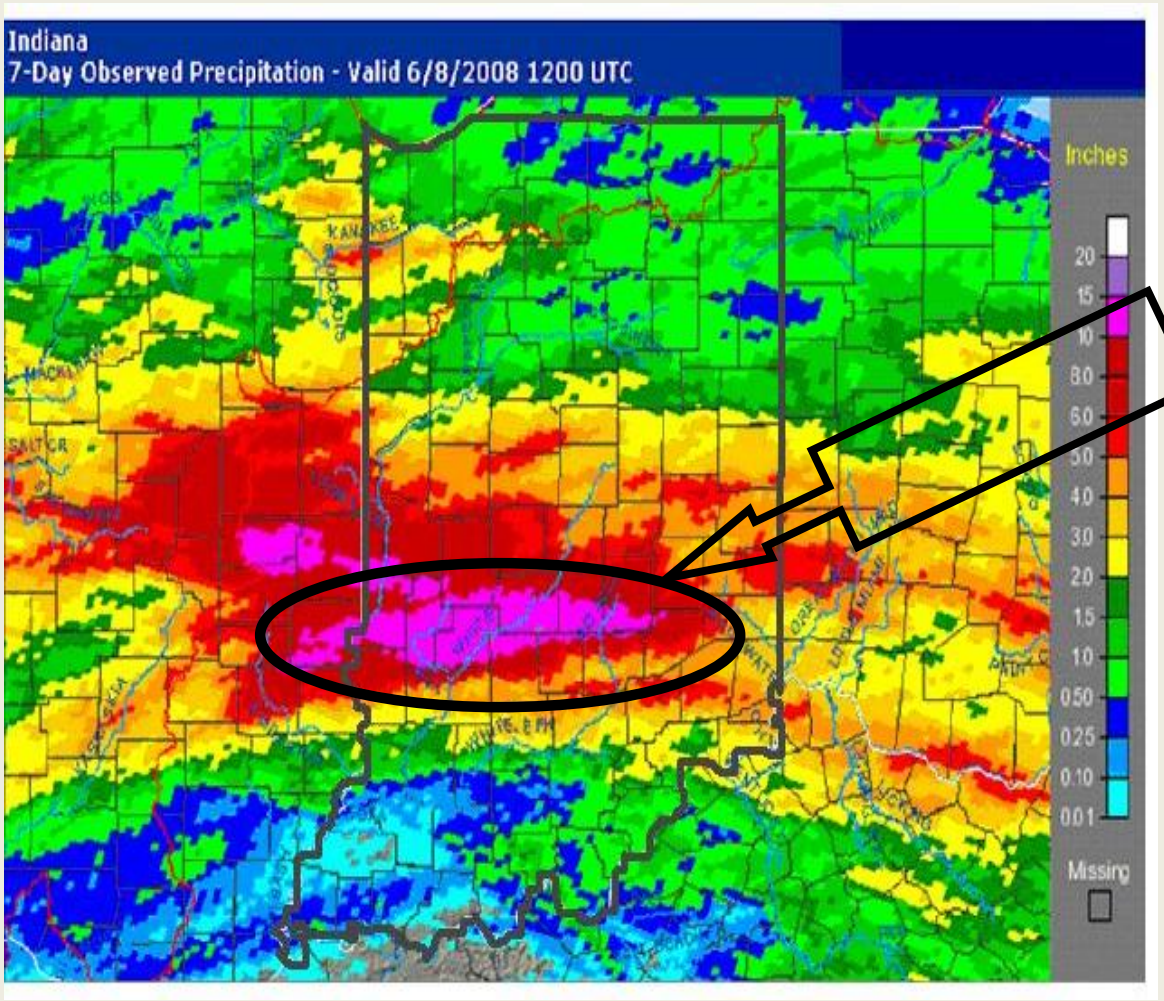
Flood Extent Mapping using RS Data

PRECIPITATION ON JUNE 7, 2008



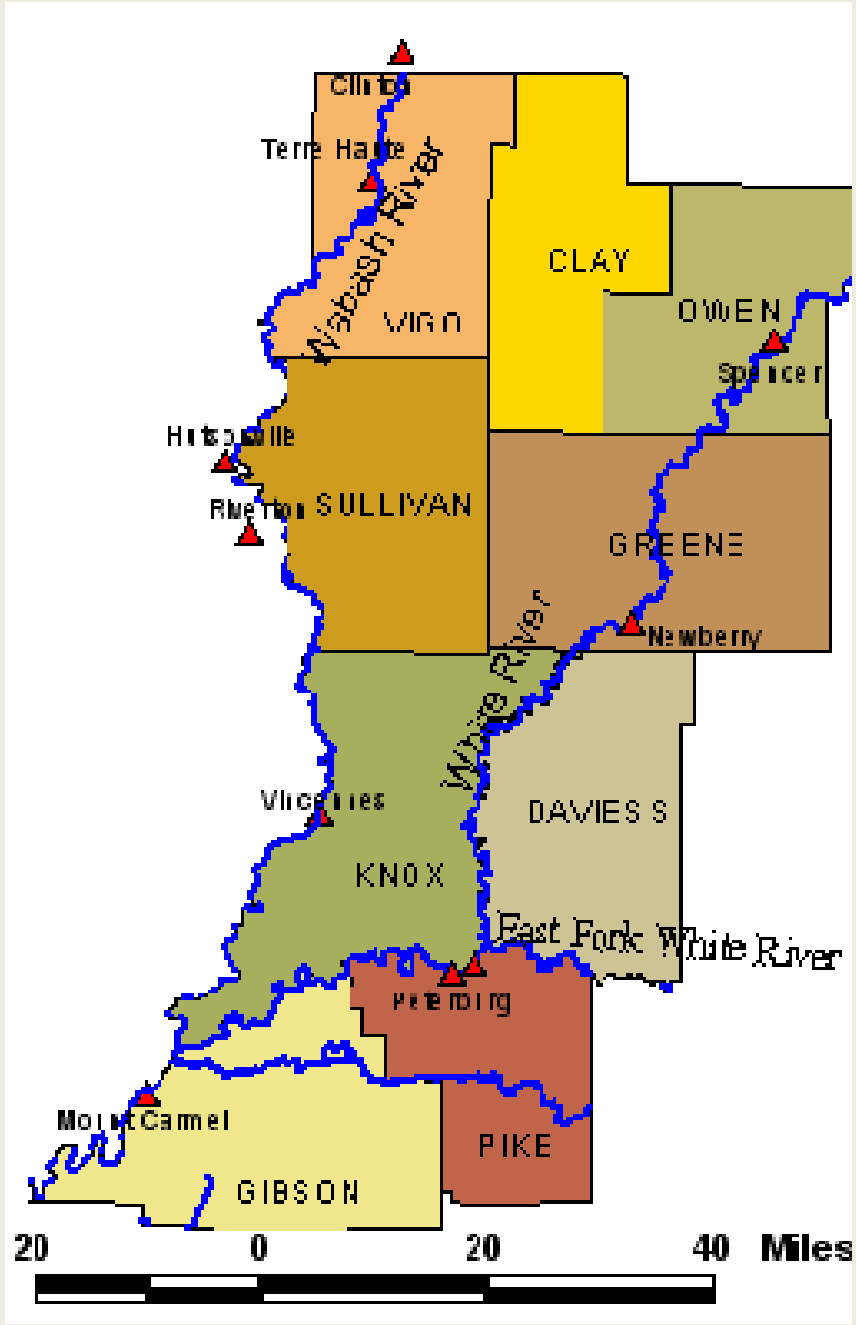
Courtesy National Weather Service

PRECIPITATION ON JUNE 1-7, 2008

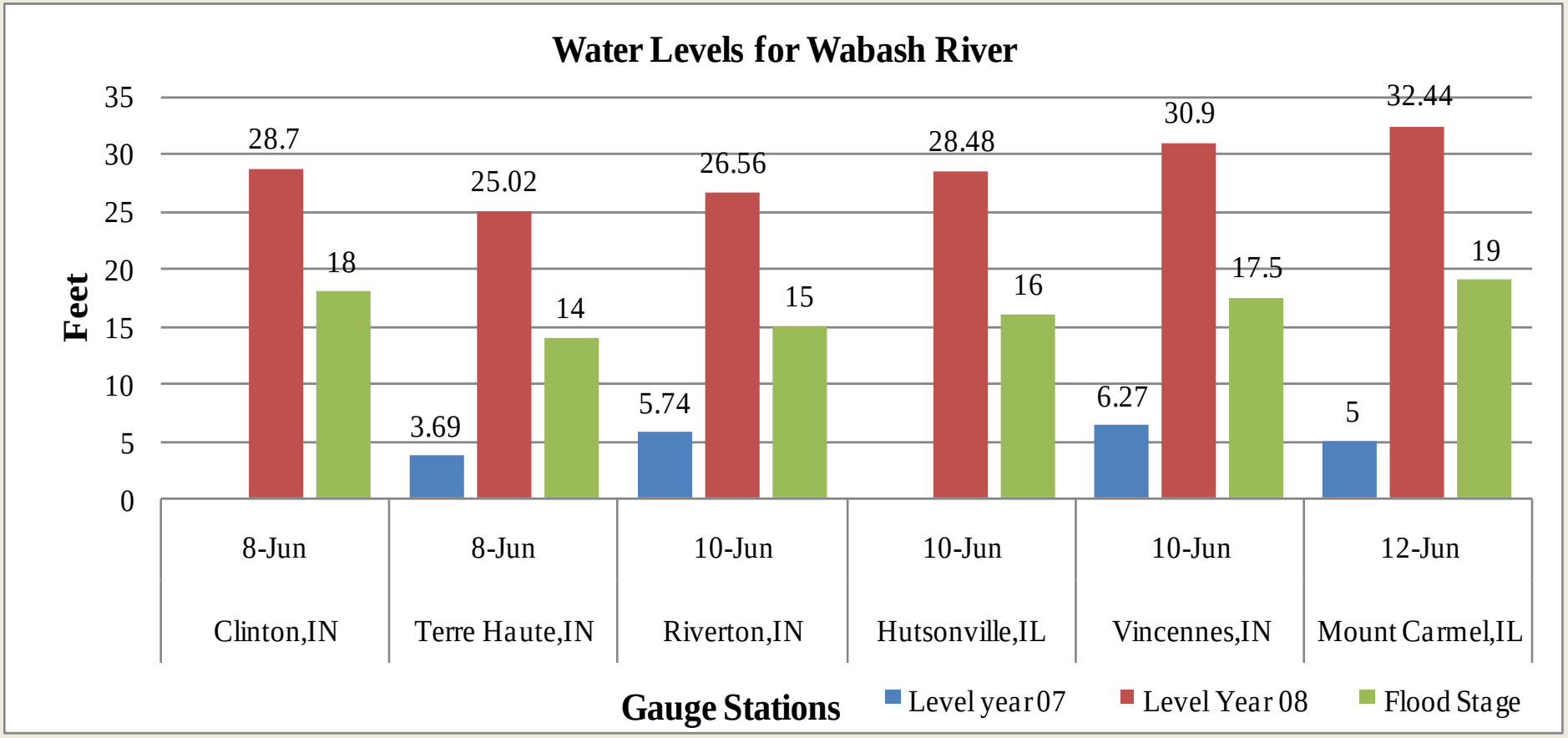


Courtesy National Weather Service

STUDY AREA-USGS GAGE STATIONS ALONG RIVERS



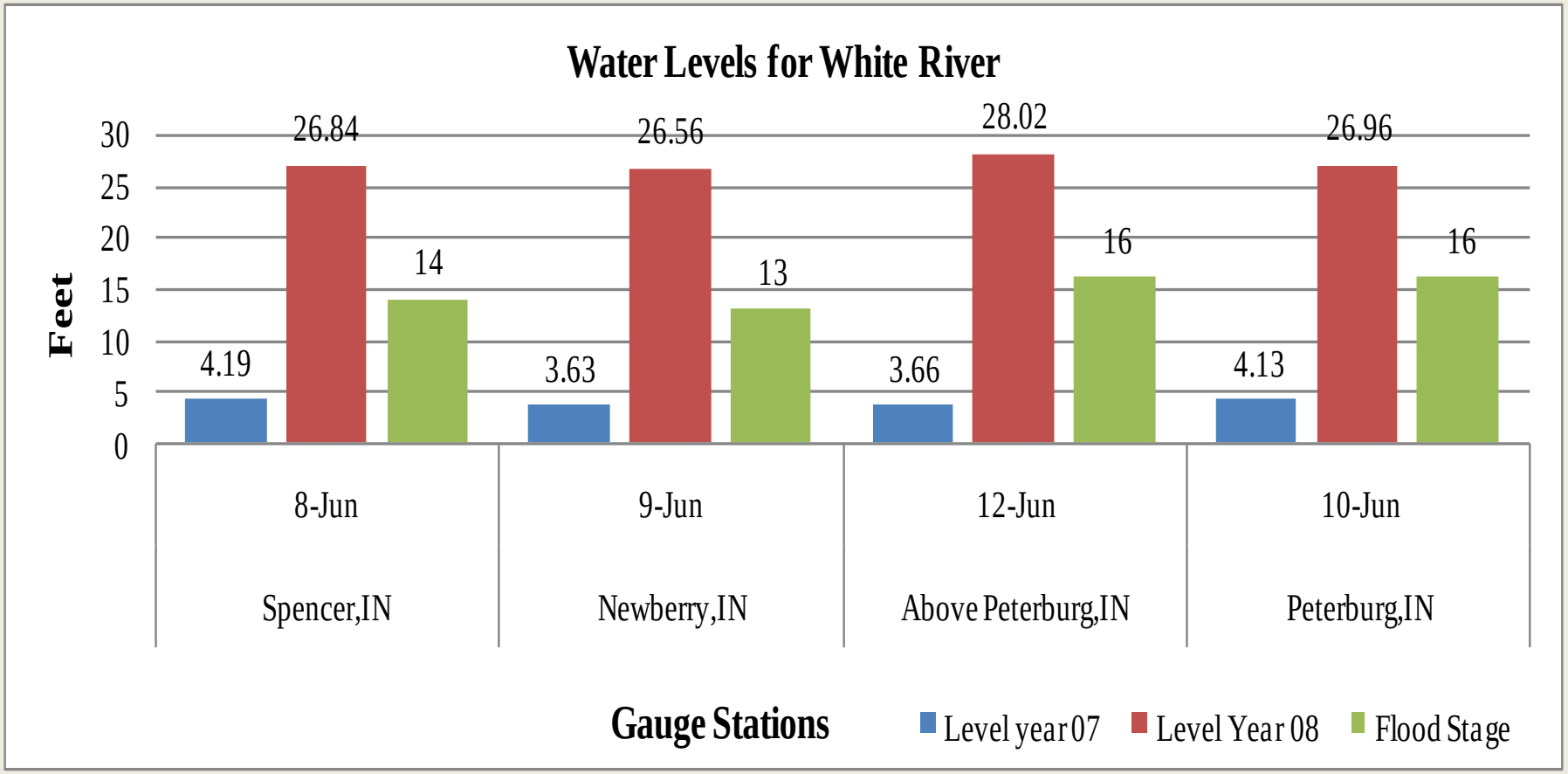
Gage Stations



- **An average of about 12 feet water level increase in about five days**

Courtesy USGS

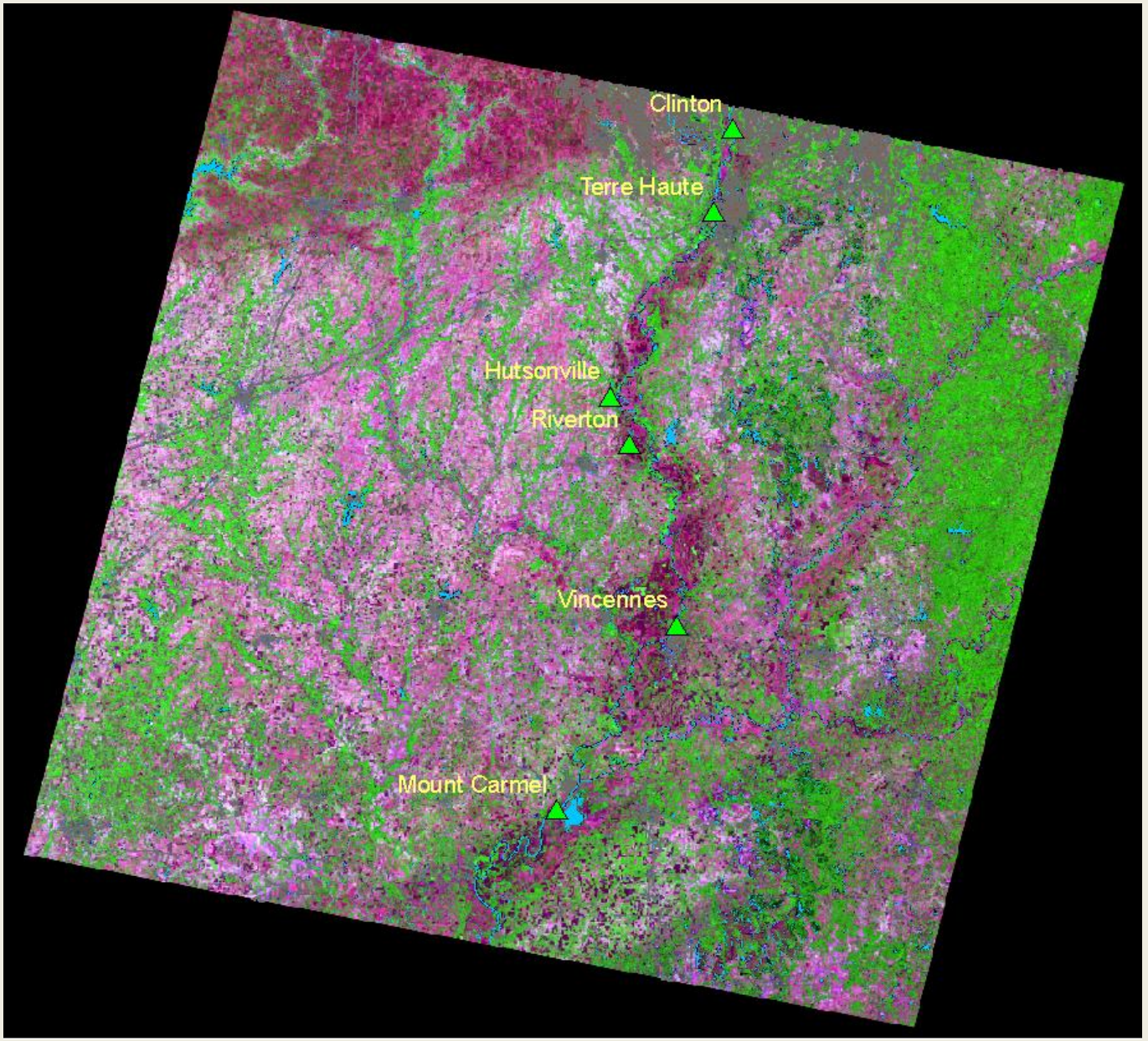
- **Highest water level recorded Mt Carmel, IL on June 14,2008- 33.24 feet (remained 8 days above flood stage)**



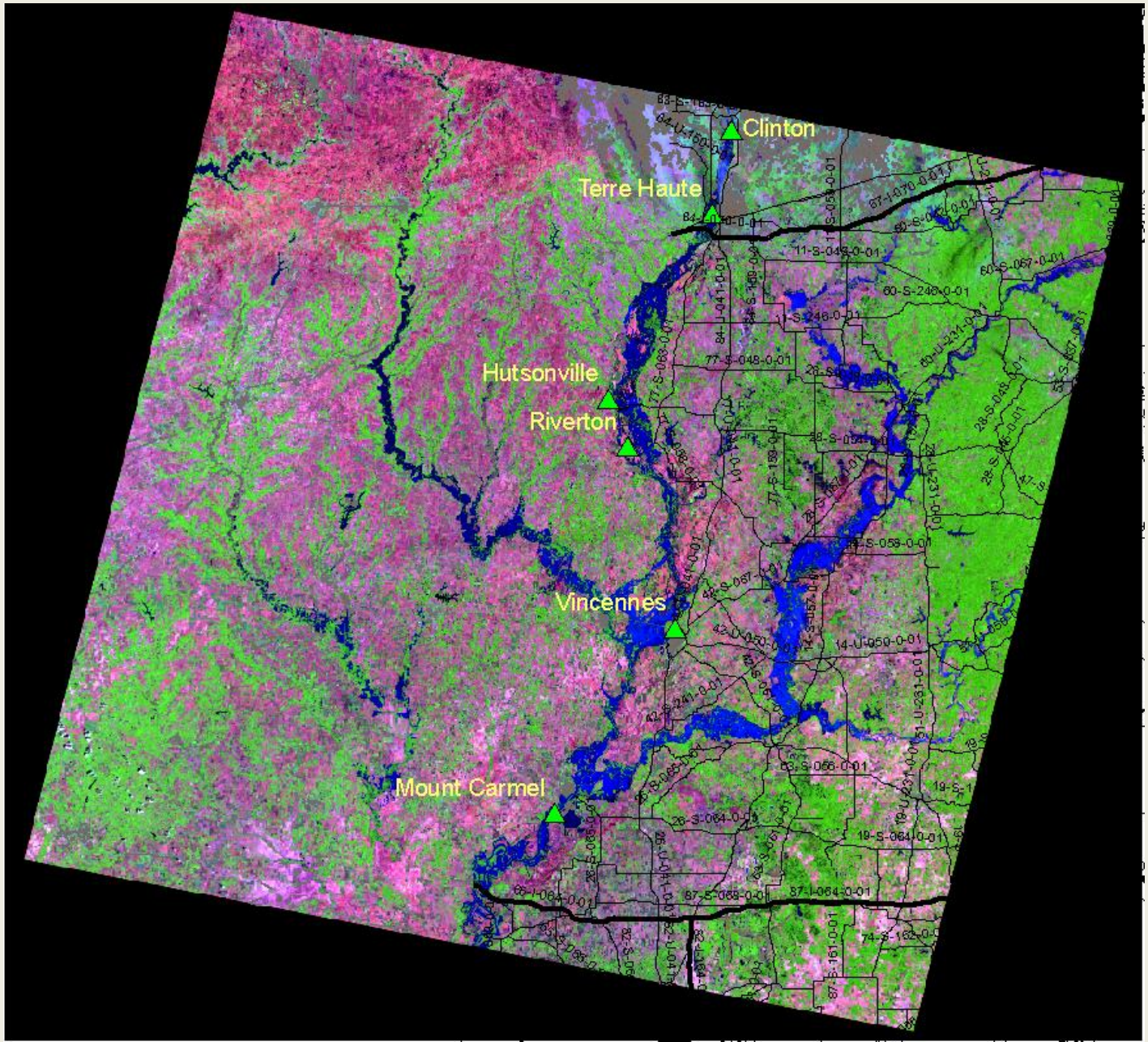
Courtesy USGS

- **An average of about 12 feet water level increase in about five days**

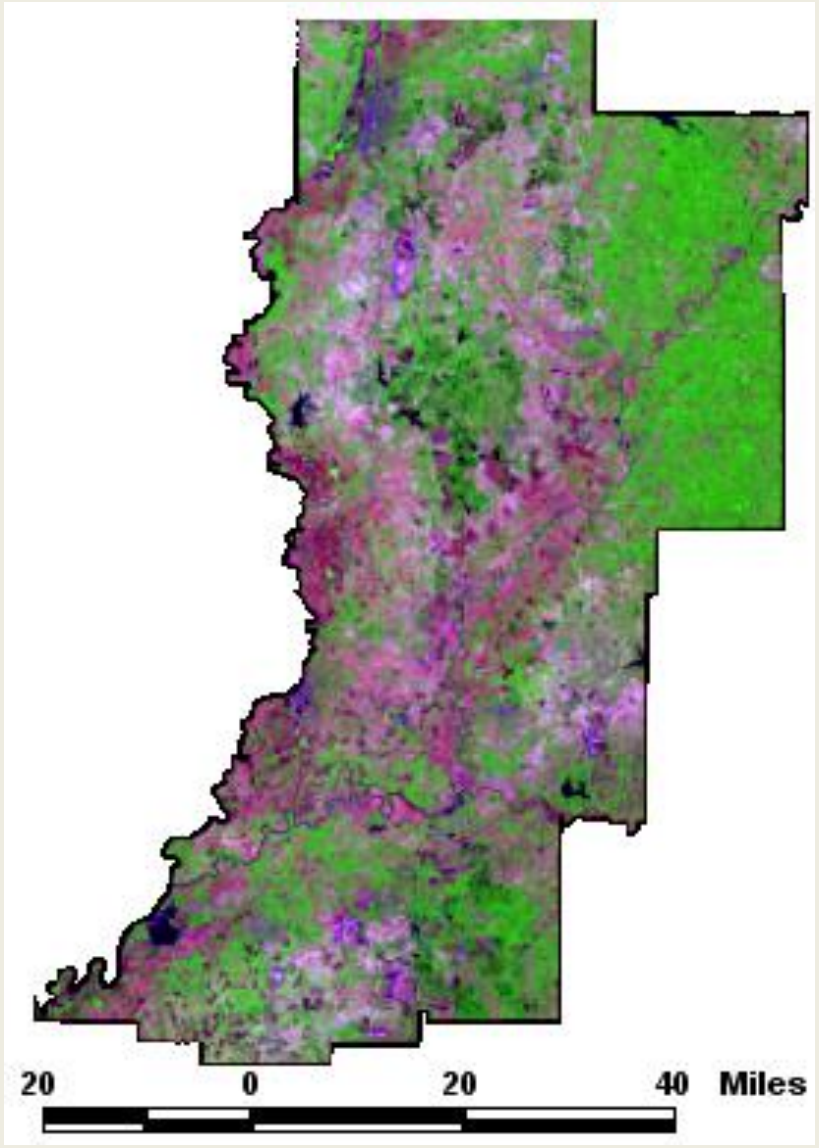
Pre Event


Clinton
Terre Haute
Hutsonville
Riverton
Vincennes
Mount Carmel

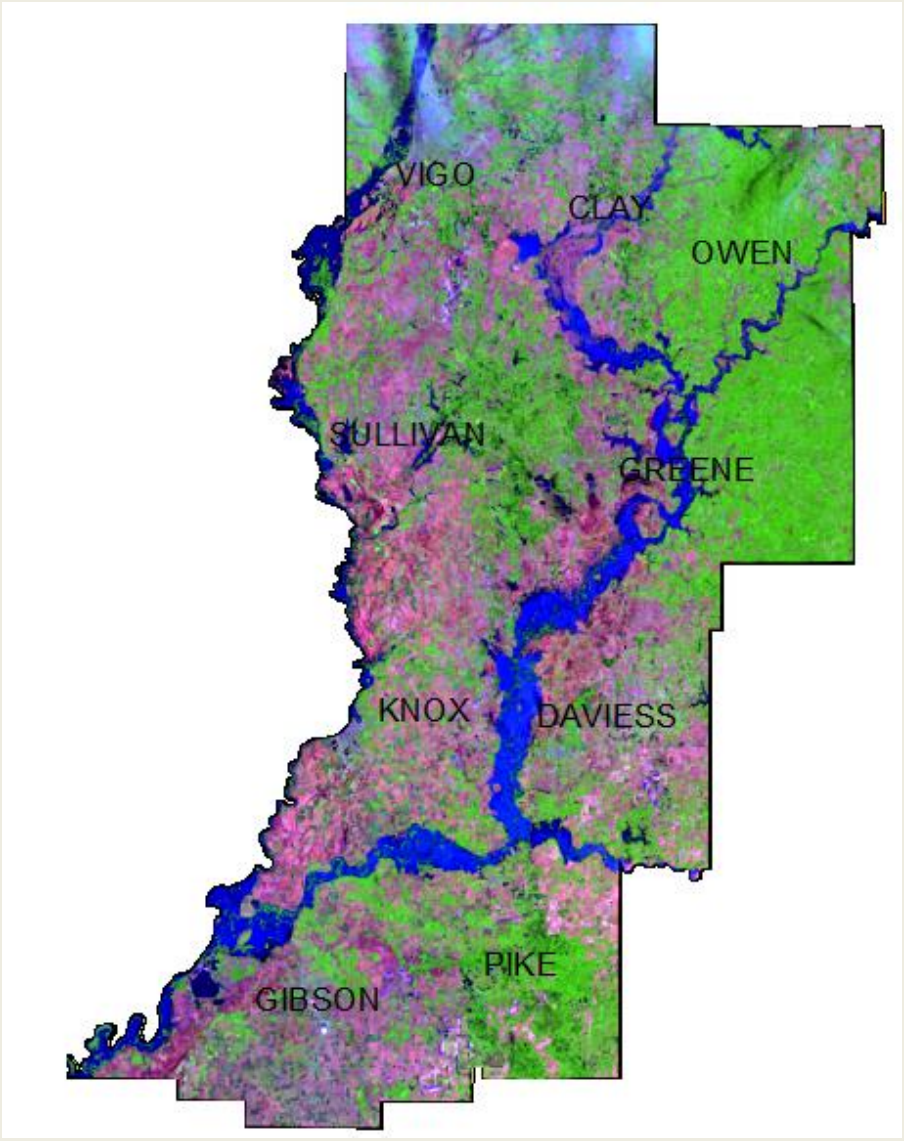
Post Event


Clinton
Terre Haute
Hutsonville
Riverton
Vincennes
Mount Carmel

STUDY AREA LANDSAT IMAGES



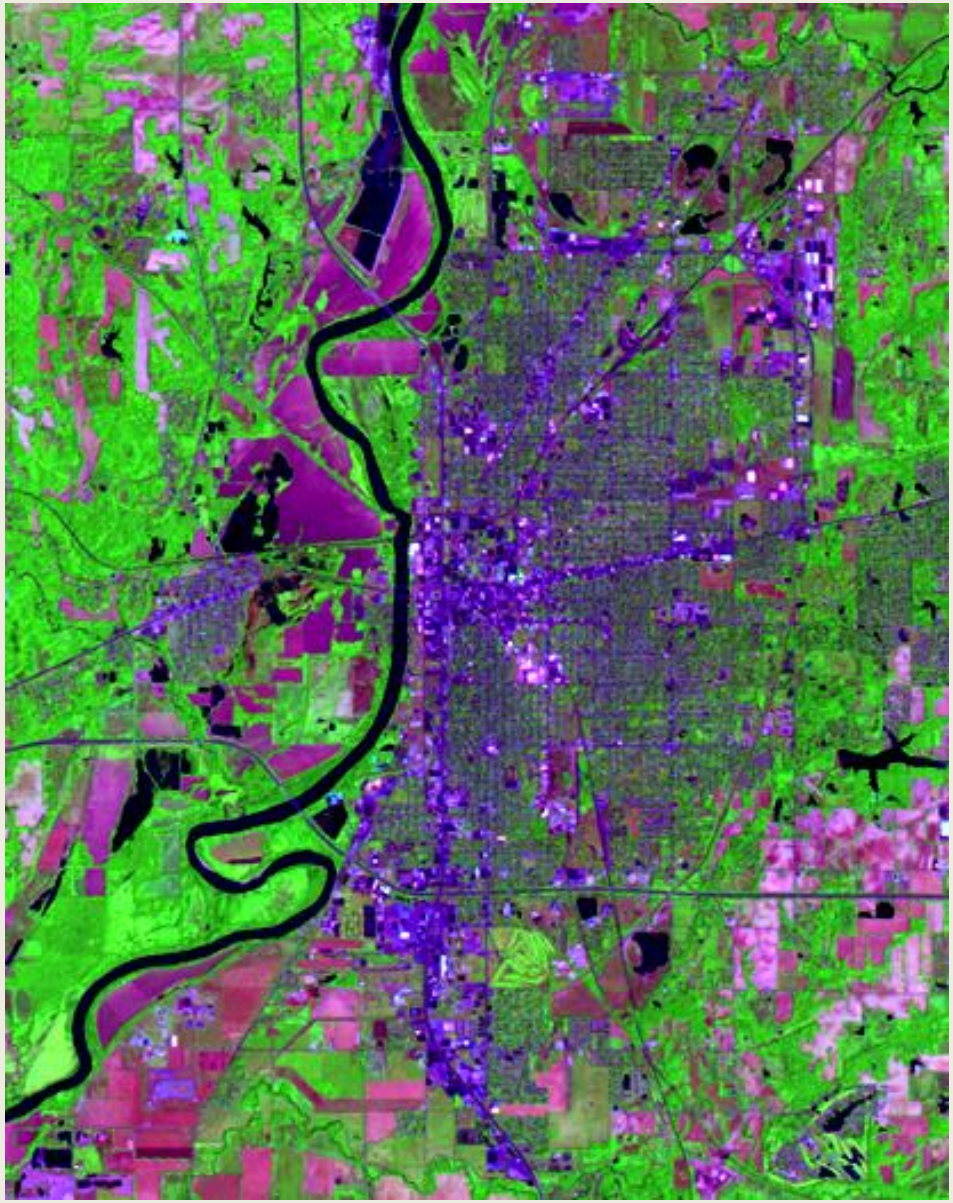
LANDSAT TM_JUNE 09, 2007



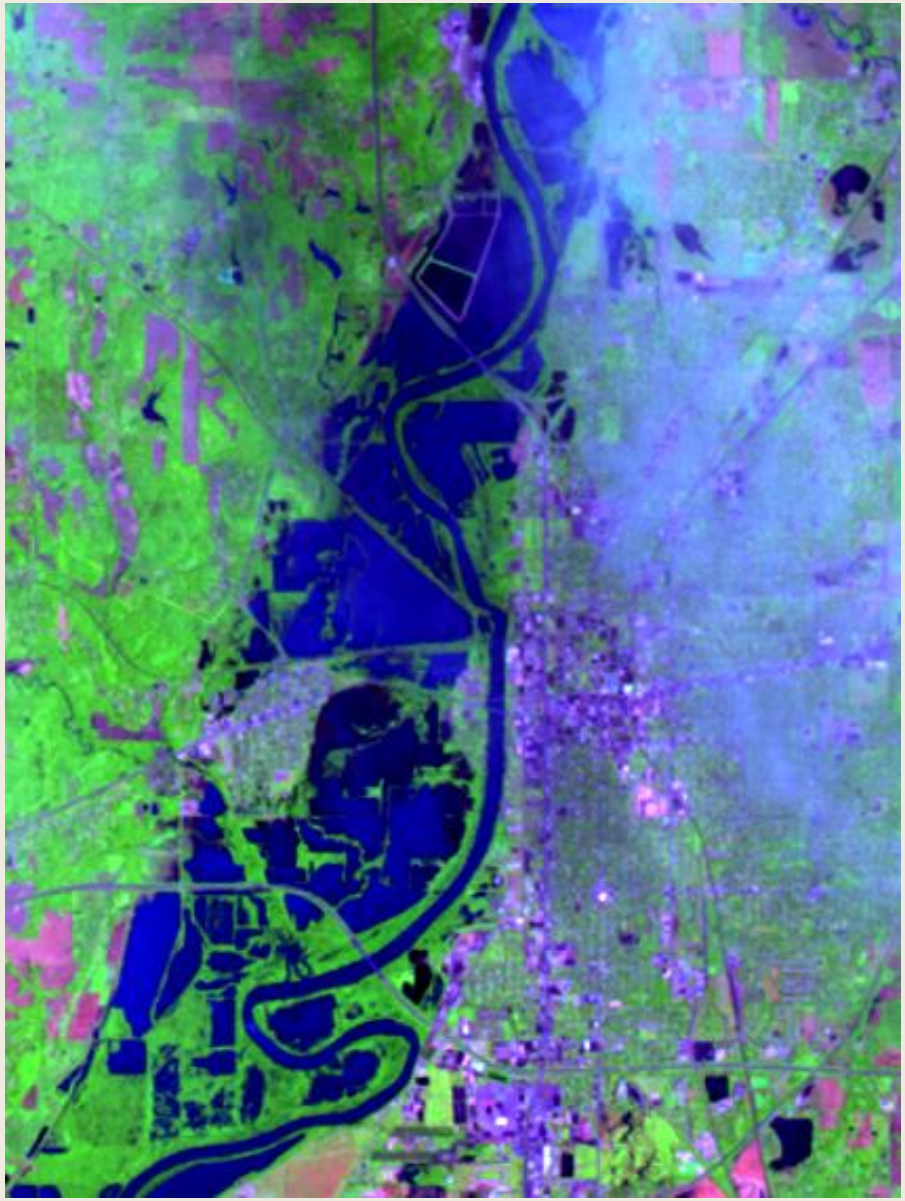
LANDSAT TM- JUNE 11,2008

Bands 5, 4, 3 displayed as red, green and blue

Pre Event-2007



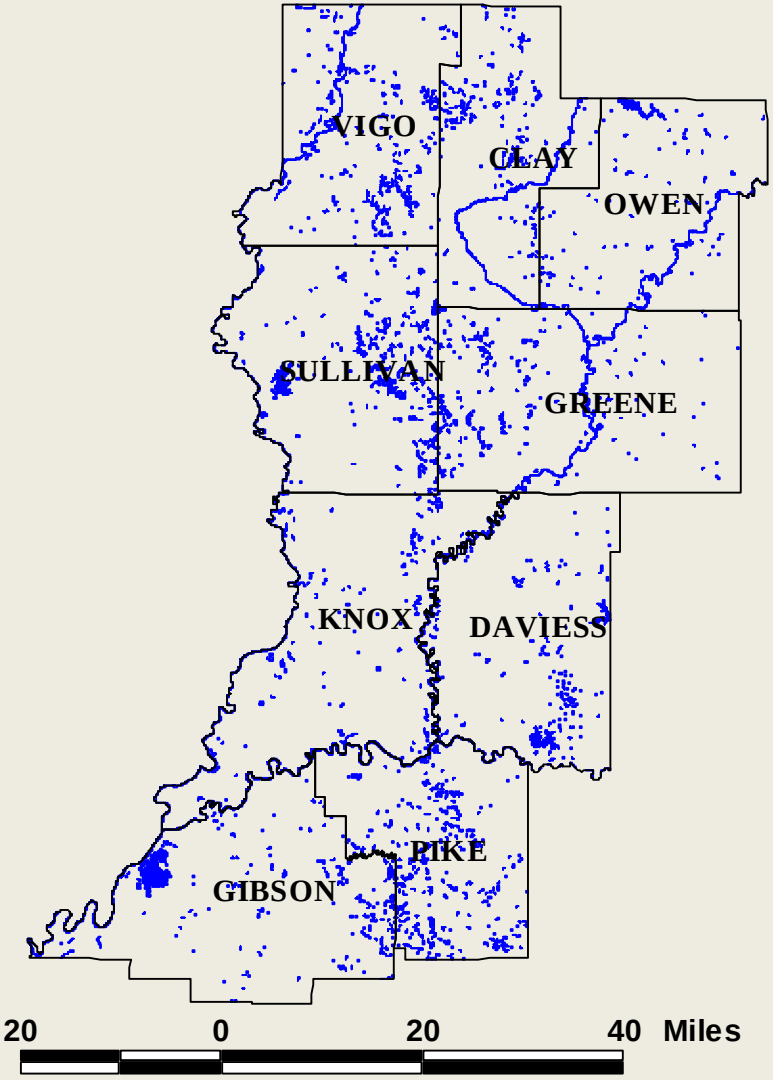
Post Event- 2008



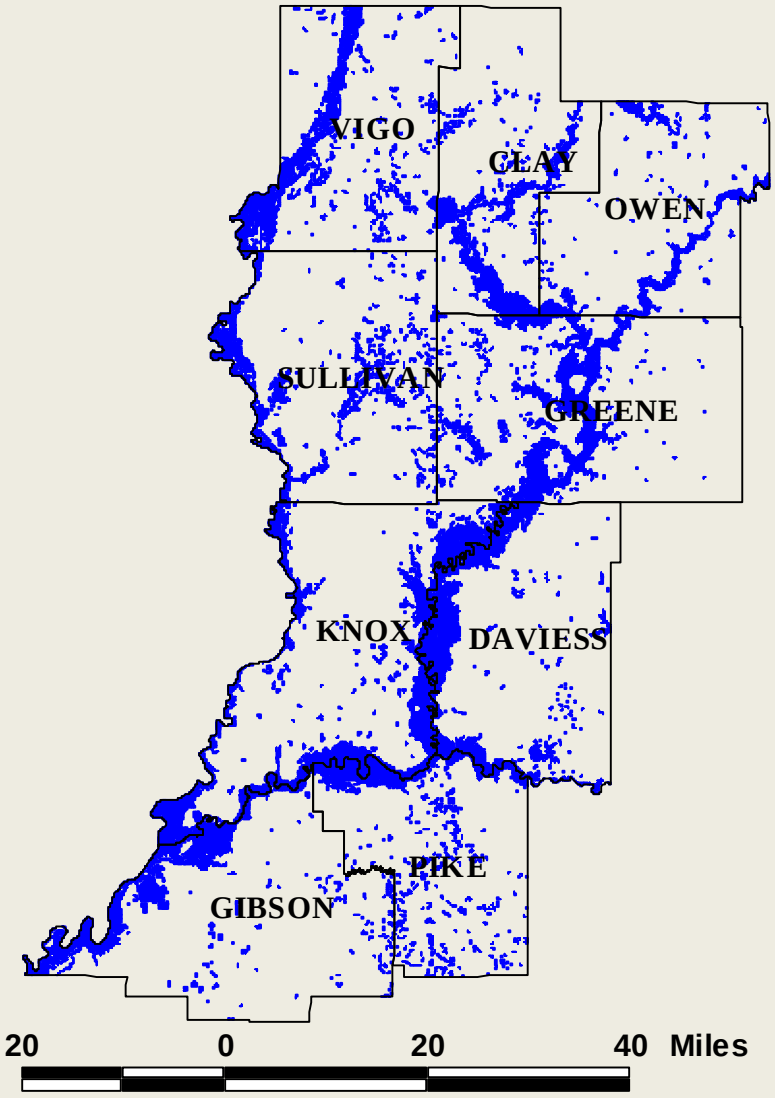
Aerial Image



CLASSIFICATION RESULTS

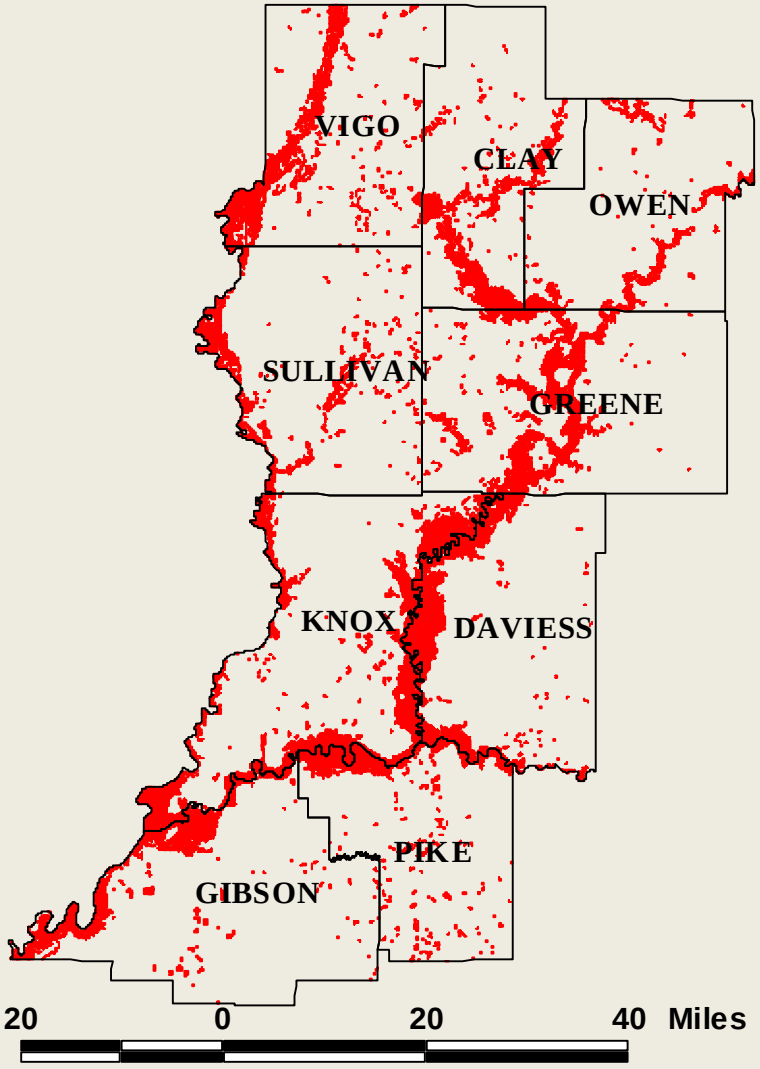


WATER_2007



WATER_2008

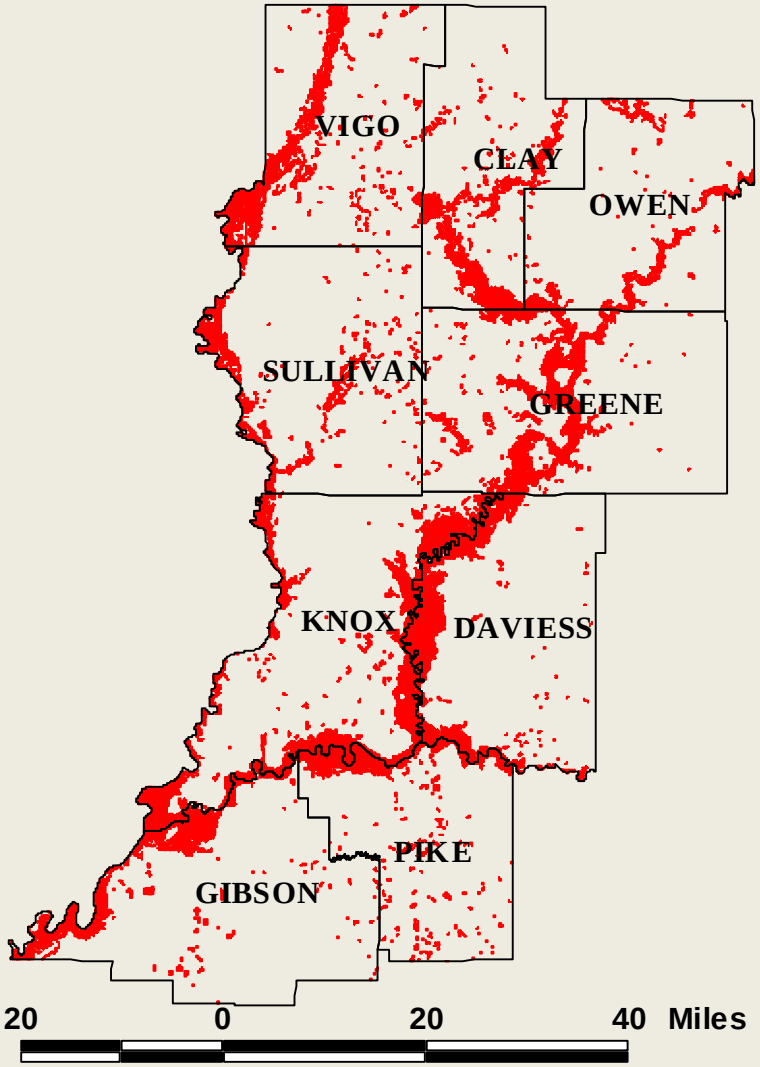
FLOOD EXTENT



Excluding 2007 water

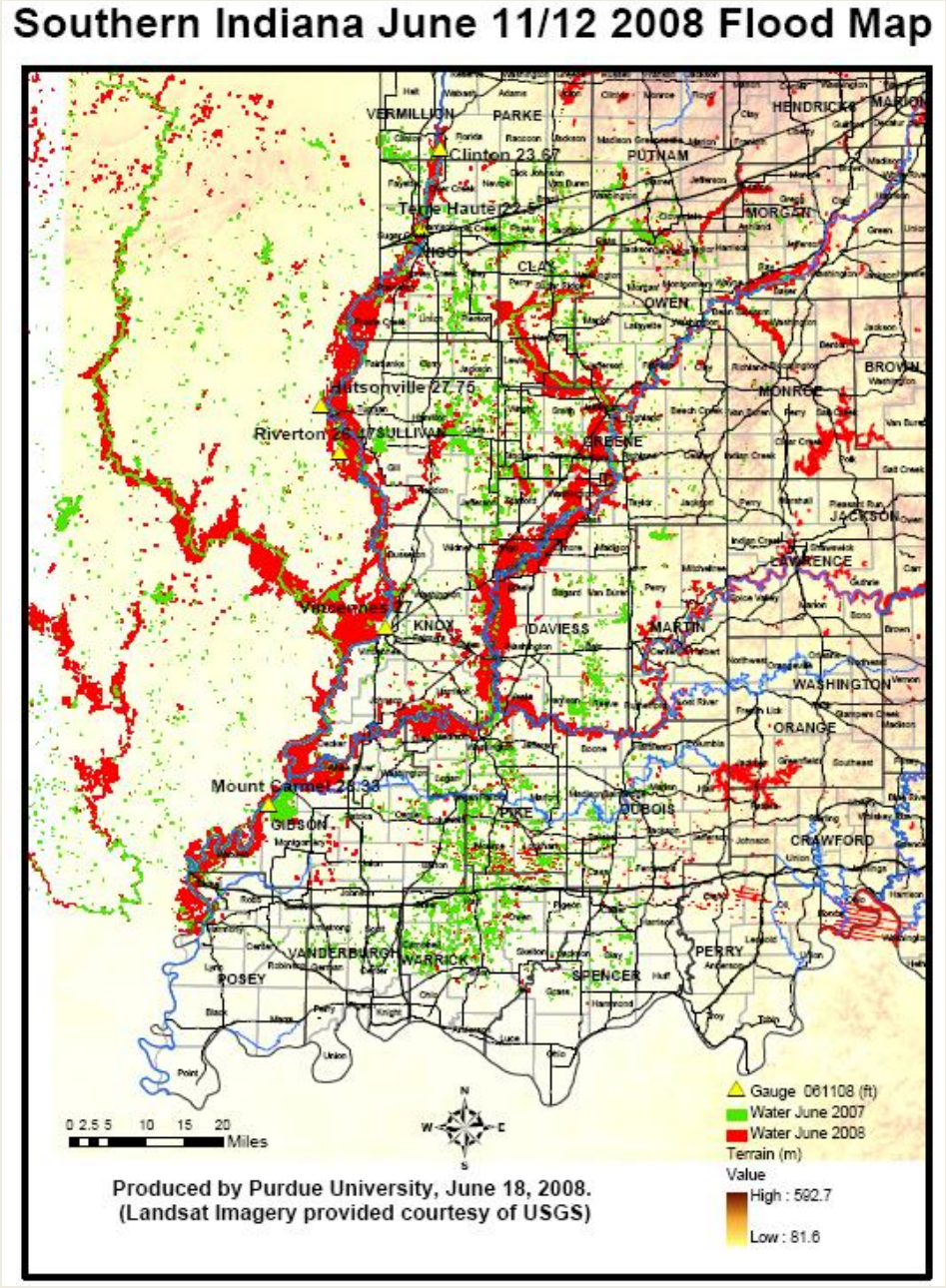
FLOOD EXTENT AND AFFECTED AREAS

County	Area (Hectares)	Flooded area (Hectares)	% Flooded area
Vigo	106224	10209	10
Clay	93237	8272	9
Owen	100317	4359	4
Sullivan	117465	6715	6
Greene	141291	14534	10
Knox	135653	20638	15
Daviess	113004	12709	11
Pike	88302	5994	7
Gibson	129159	9940	8



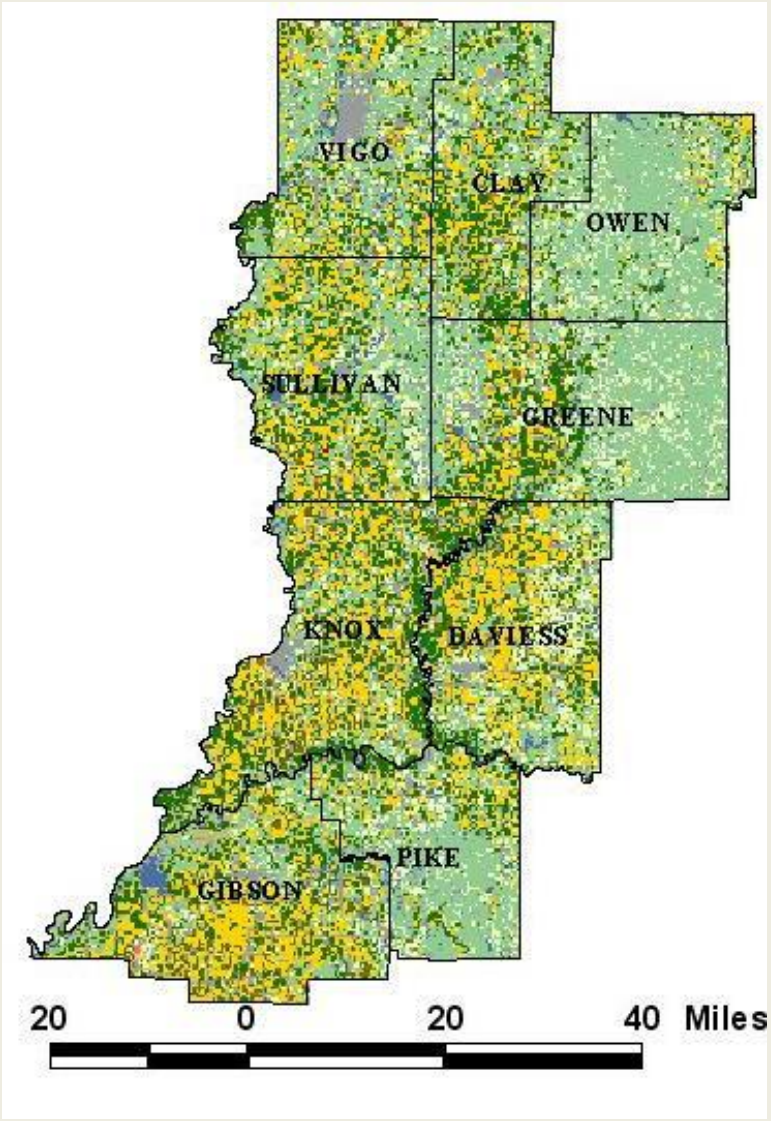
Total Flood area = 93,000 Hectares
Excluding 2007 water

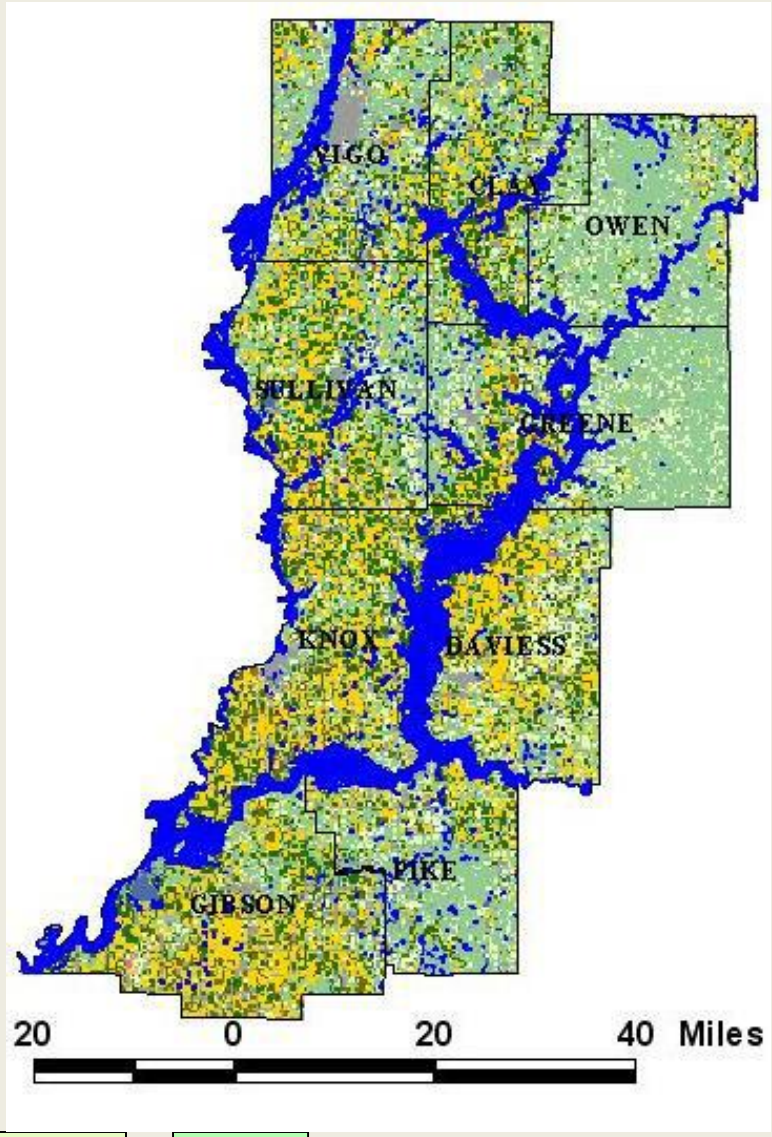
Southern Indiana June 11/12 2008 Flood Map



https://engineering.purdue.edu/CE/Academics/Groups/Geomatics/floodmaps

USDA CROPS DATA-2008





Corn | Soybean | Wheat | Pasture | Woods

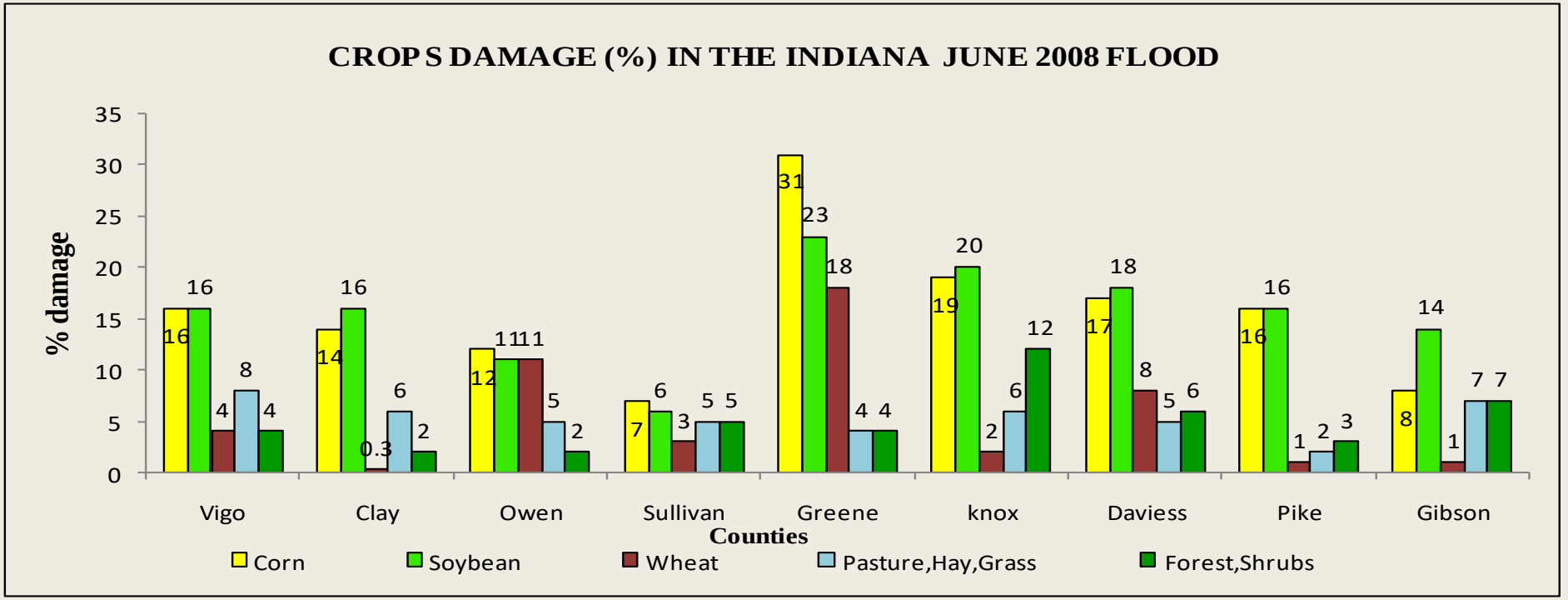
USDA Crops 2008

USDA Crops With 2008 Water Layer

CROP DAMAGE SUMMARY-NINE COUNTIES 2008

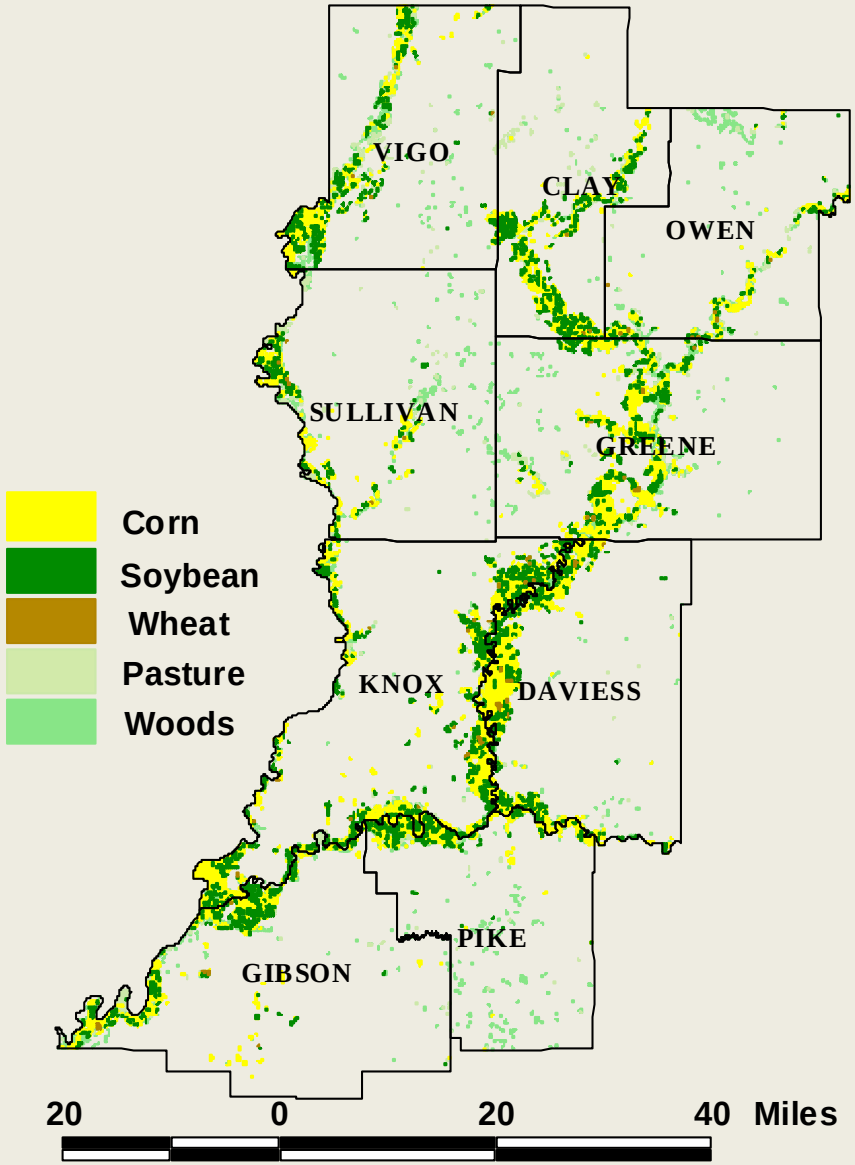
	Crops area under flood (Hectare)				
County	Corn	Soybean	Wheat	Pasture, Hay	Forest, Shrubs
Vigo	1823	3911	49	680	1675
Clay	1586	4058	23	208	1025
Owen	393	1336	31	510	1539
Sullivan	752	1994	18	647	1633
Greene	1607	7315	118	652	2061
Knox	1852	10496	63	526	2604
Daviess	1682	6544	38	297	1359
Pike	252	3265	2	249	1071
Gibson	780	2543	5	293	1362
Total Area	10728	41462	347	4062	14329

SUMMARY-NINE COUNTIES

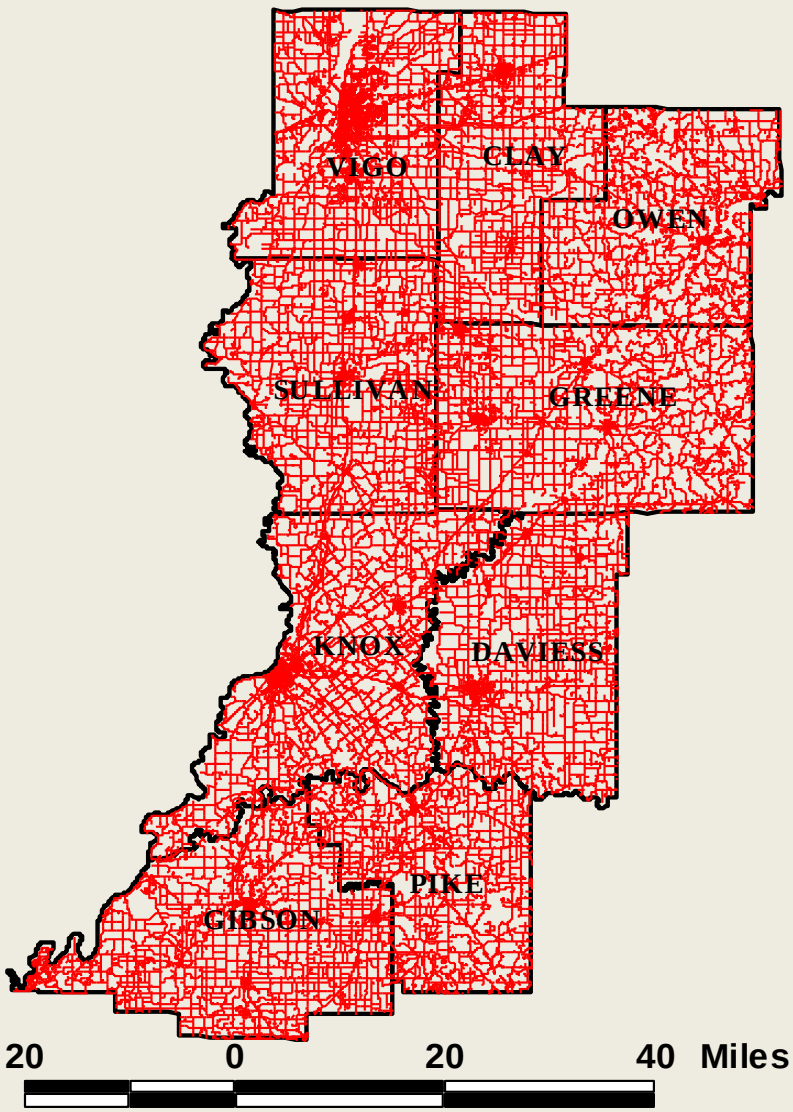


An average of about 15% of corn and soybeans areas affected by floods

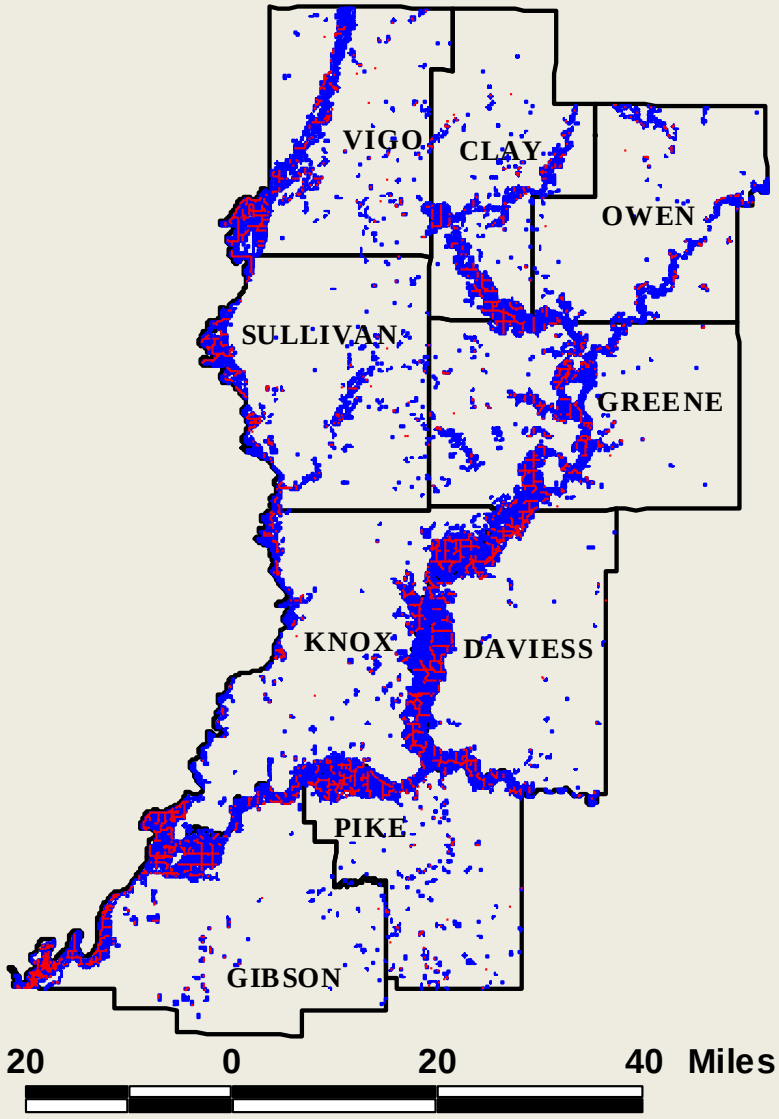
CROPS DAMAGE MAP



FLOOD AFFECTED ROADS

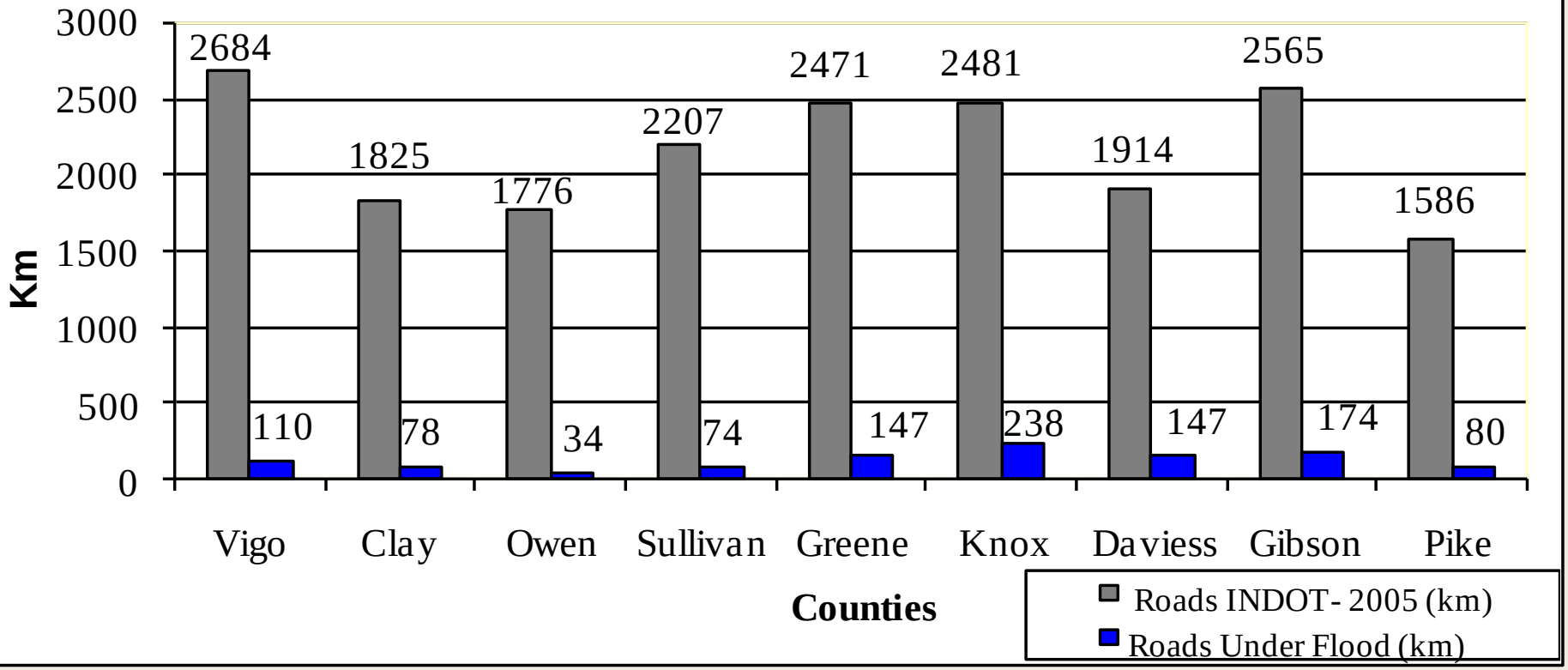


INDOT ROADS 2005



FLOOD AFFECTED ROADS

FLOOD AFFECTED ROADS

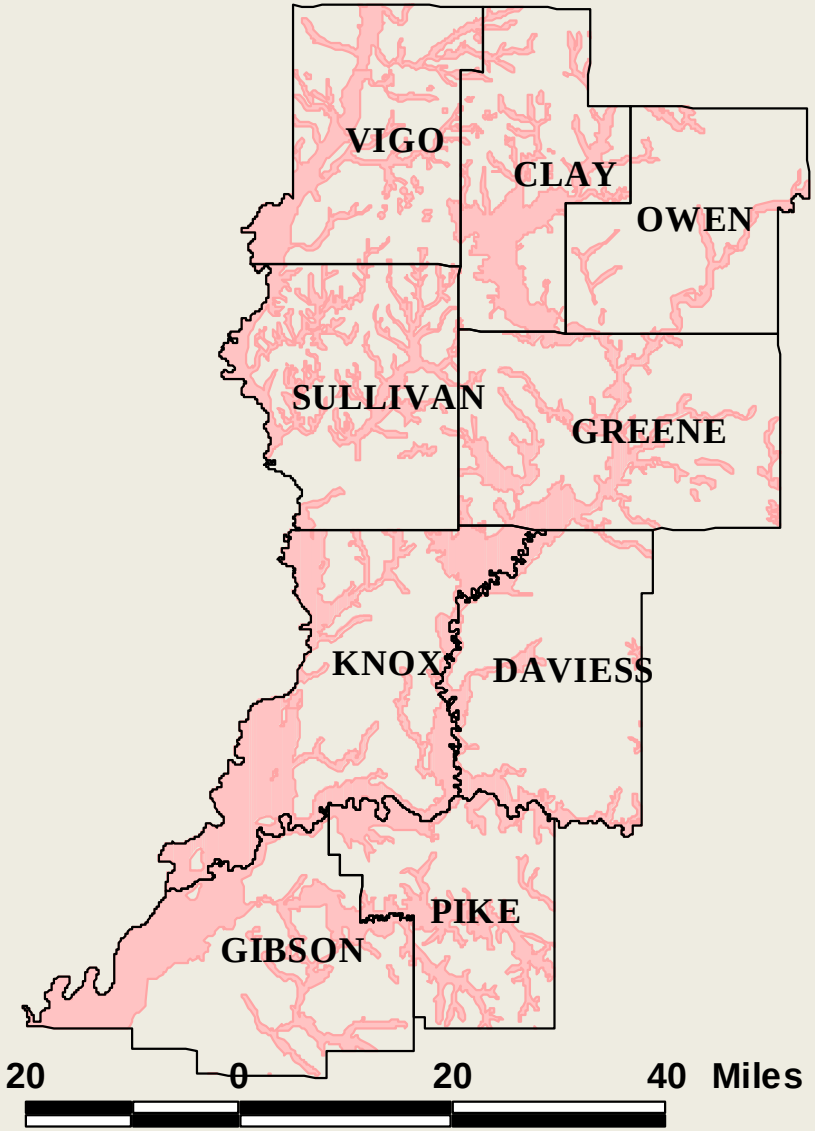


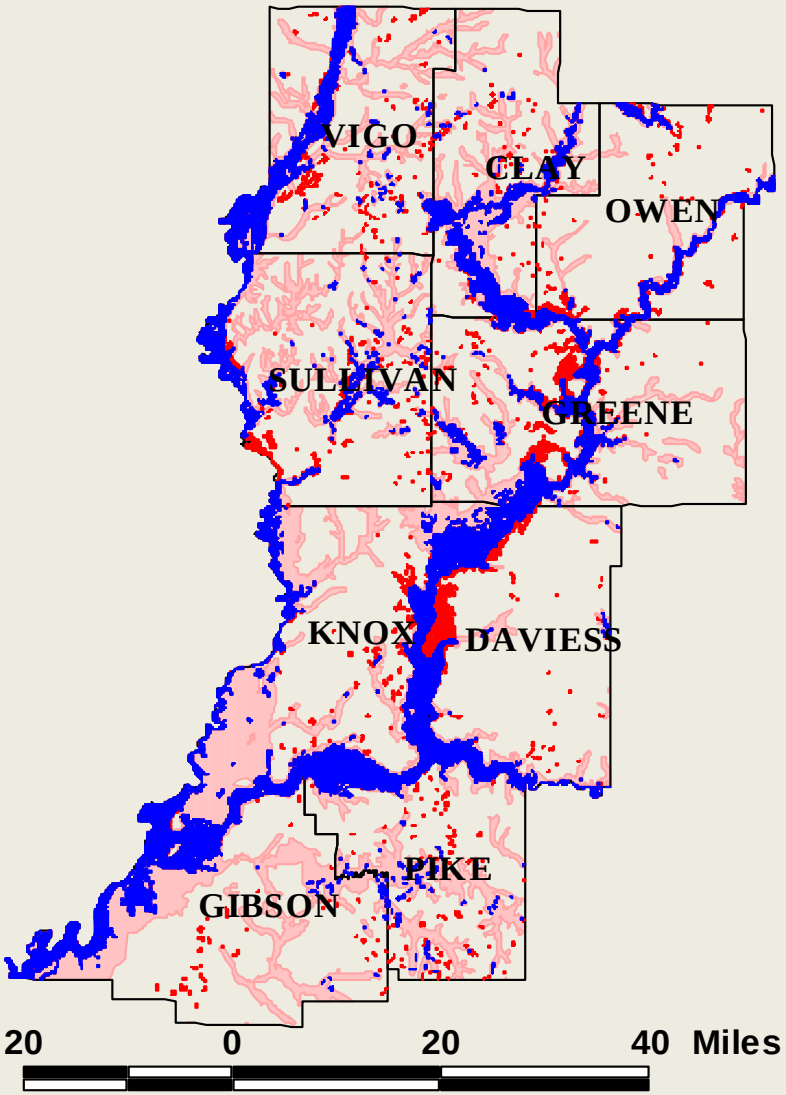
INDOT Road Closures/Restrictions Due to Flooding

Road	Status	District	County	Direction(s)	Starting Location	Ending Location
SR 42	Closed	Crawfordsville	CLAY	both	Corey Rd	CR 300 W-Clay
SR 246	Closed	Crawfordsville	VIGO	both	All Street	Dickerson St
SR 42	Closed	Crawfordsville	OWEN	both	US 231	SR 59
SR 46	Closed	Seymour	Owen	Both	SR 67	SR 246
SR 144	Closed	Seymour	Morgan	Both	SR 37	SR 67
SR 157	Closed	Vincennes	Greene	Both	SR 54	SR 67
SR 58	Closed	Vincennes	Daviess	Both	SR 67	SR 57
SR 57	Closed	Vincennes	Greene	Both	SR 58	SR 54
SR 257	Closed	Vincennes	Daviess	Both	US 50	SR 356
SR 66	Restricted	Vincennes	Posey	Both	IL Line	SR 69

http://www.in.gov/dot/gis/slo/RoadCloseText.pdf

FLOODPLAINS GENERAL





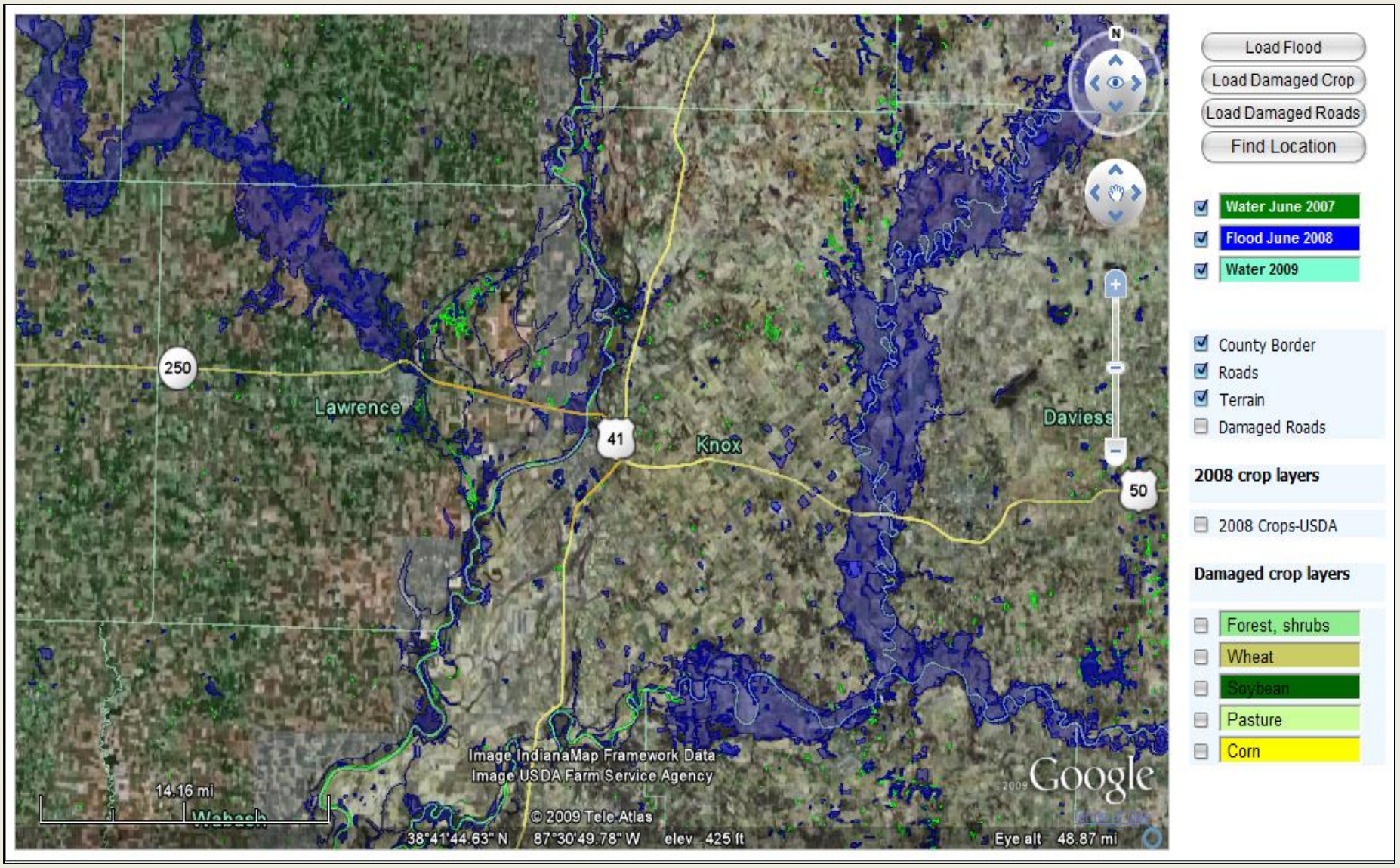
FLOODPLAINS FLOOD WATER IN FLOODPLAINS FLOOD WATER OUT OF FLOODPLAINS

93000 Hectares

6000 Hectares

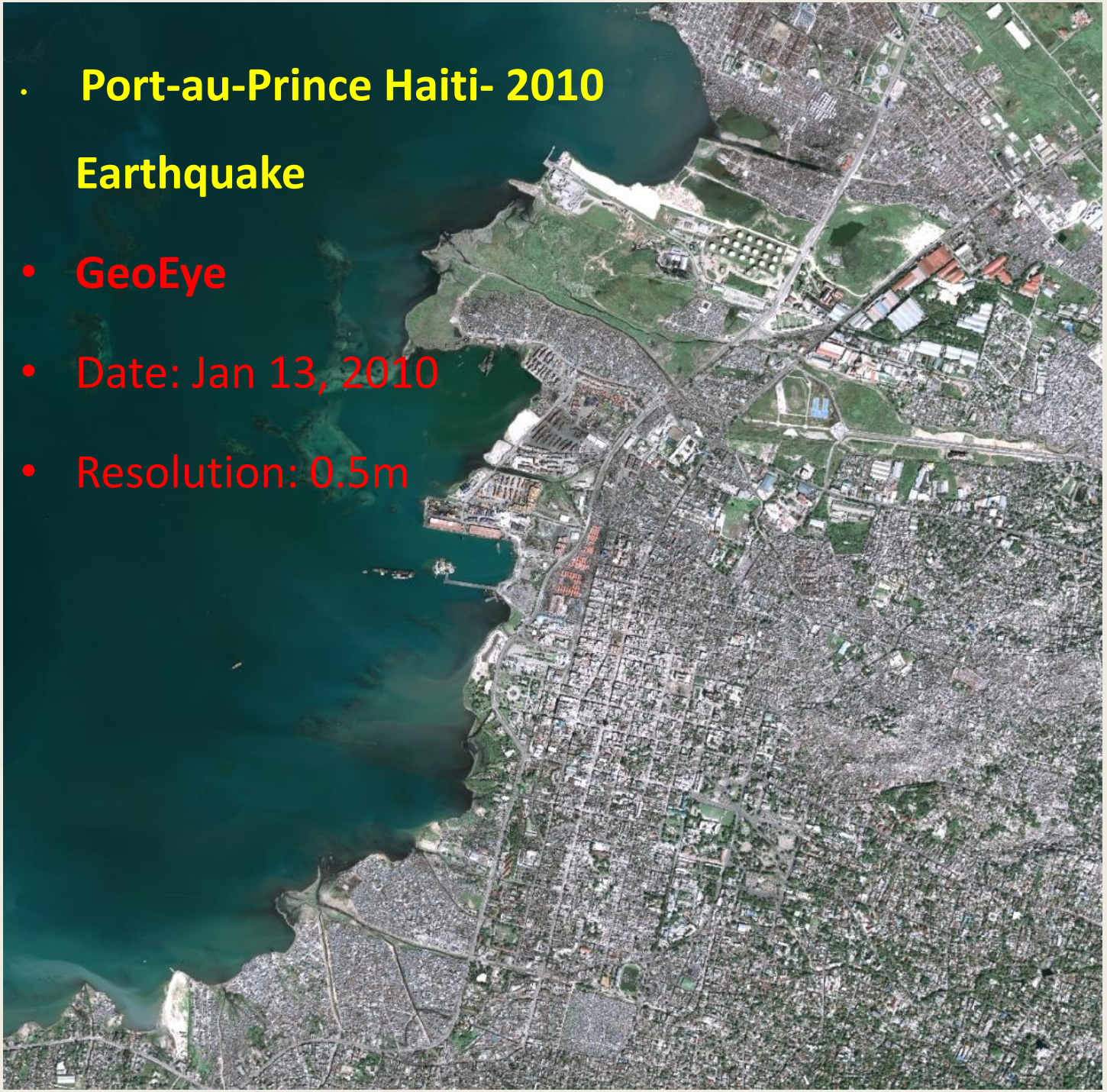
Web Mapping

Web interface



https://engineering.purdue.edu/CE/Academics/Groups/Geomatics/floodmaps




Port-au-Prince Haiti- 2010 Earthquake
GeoEye
Date: Jan 13, 2010
Resolution: 0.5m


• 1-m LiDAR DSM
• Date: Jan 21/22, 2010
• World Bank, RIT,

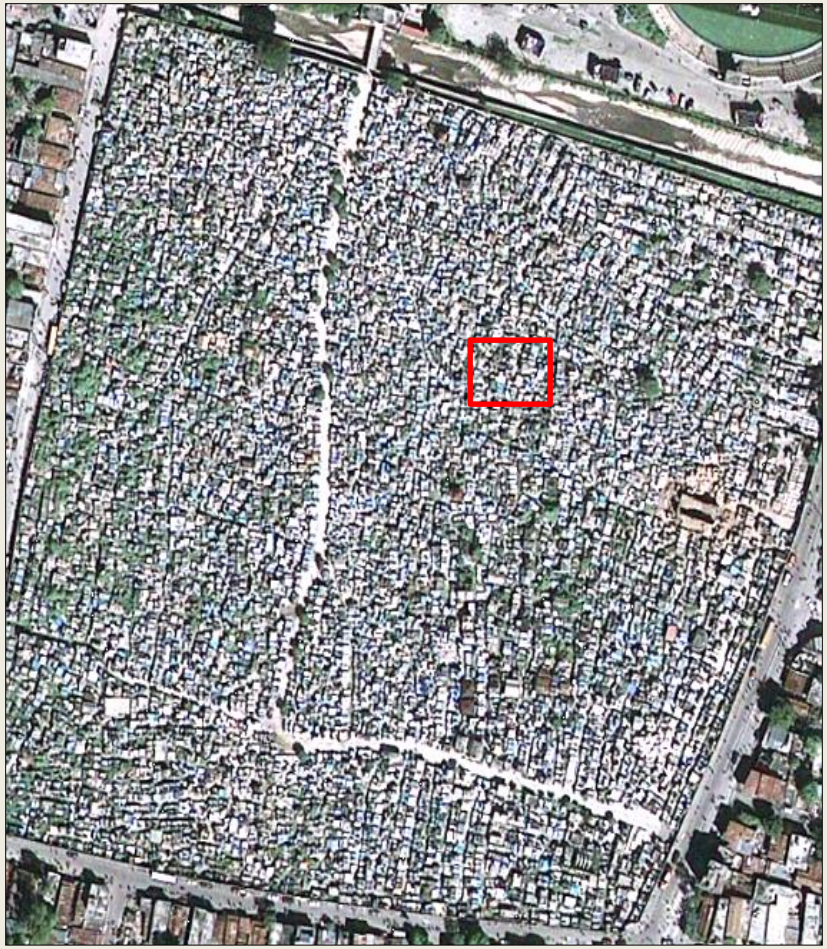
Urban complexity-Port au Prince








Types of Buildings

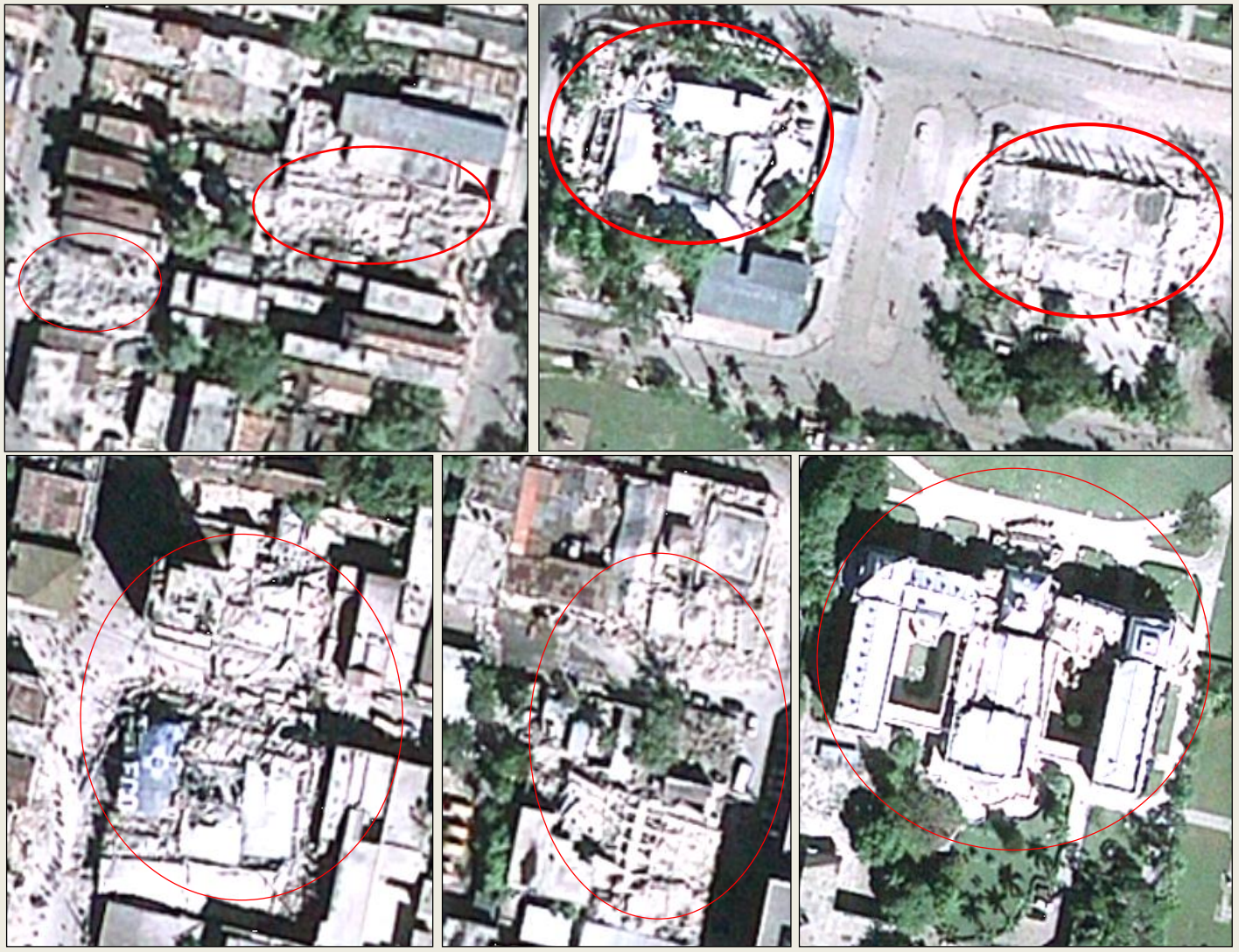


GeoEye-1 (0.5 m)



Aerial Image (3 inch)

Damaged Buildings



Complete and Partial Collapse

Damaged Buildings



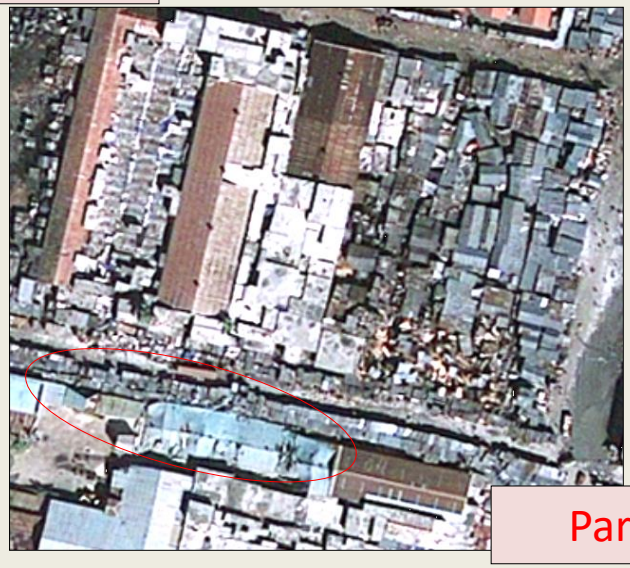


Roof collapse





Side walls collapse



Partial collapse

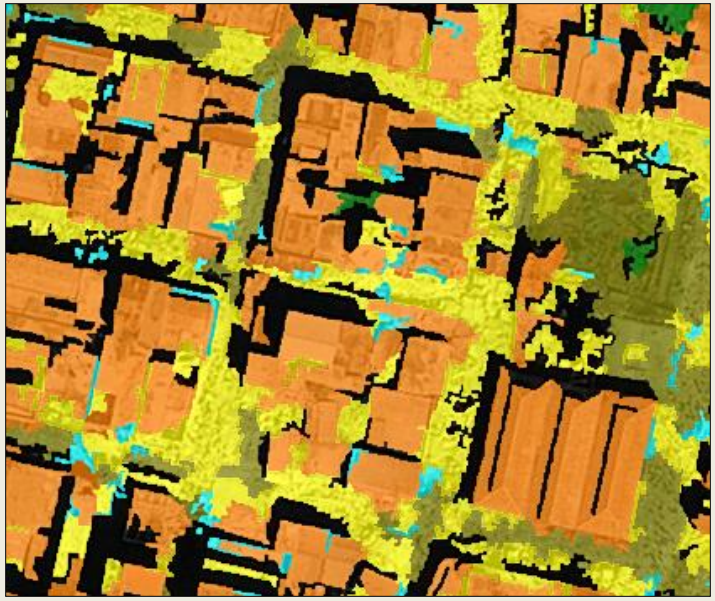
Rubble Mapping



Rubble

Roads and Streets-Rubble/Debris







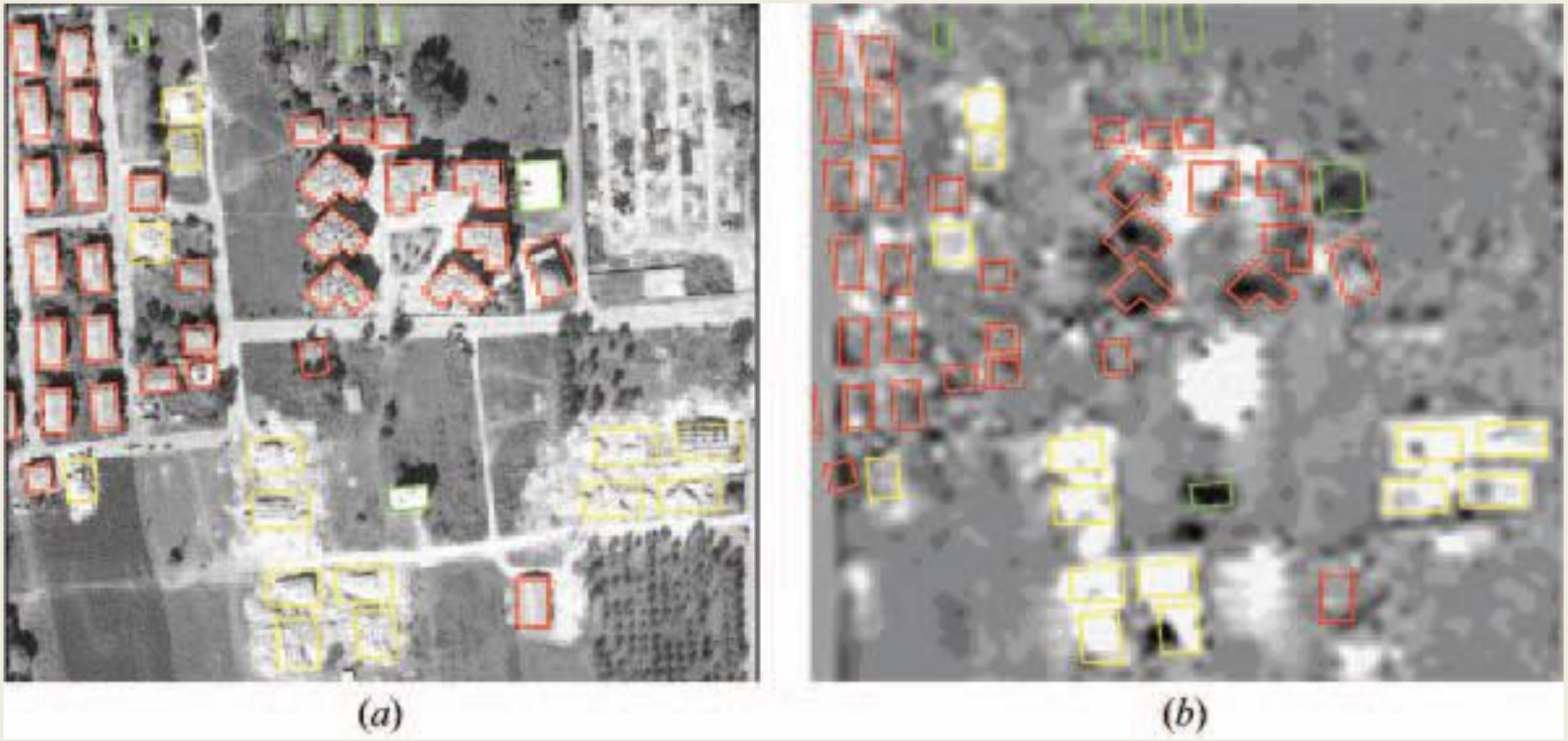
Rubble/debris

Pre- and post-earthquake aerial images



(a) (b)

the building vector polygons overlaid



(*a*) (*b*)

Conclusions

- Timely availability of Remote Sensing data helps
  - To quickly map and visualize the disaster impact
  - To assess damage
  - To plan remedial measures
- Availability of flood extent maps on internet help allocating /routing resources